



A Knowledge Distillation Framework for Guava Disease Classification Using Lightweight Deep Learning Models

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The undergraduate thesis titled “A Knowledge Distillation Framework for Guava Disease Classification Using Lightweight Deep Learning Models” submitted by ASRAR TAMIM (ID: 221009112) has been accepted as satisfactory in fulfillment of the requirement for the degree of Bachelor of Science (B.Sc.) in Computer Science and Engineering (CSE) to be awarded by East Delta University.

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Declaration

It is hereby declared that the work contained in this thesis is original. The information derived from the literature or work has been duly acknowledged and presented in the reference section. No part of this thesis has been submitted elsewhere for the degree, diploma or other similar title of recognition.

Date: 30 DECEMBER ,2025.

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LIST OF ABBREVIATIONS

CNN	: Convolutional Neural Network
KD	: Knowledge Distillation
MTKD	: Multi-Teacher Knowledge Distillation
SD	: Self-Distillation
VGG16	: Visual Geometry Group 16-layer Network
VGG19	: Visual Geometry Group 19-layer Network
ResNet50	: Residual Network with 50 Layers
Xception	: Extreme Inception Network
InceptionV3	: Inception Version 3
InceptionResNetV2	: Inception-Residual Network Version 2
MobileNet	: Mobile Neural Network Architecture

ABSTRACT

This research presents an advanced Knowledge Distillation (KD) framework designed to enhance the performance of lightweight student models for image classification tasks in resource-constrained environments. To leverage the powerful representation capabilities of deep learning models, five pretrained teacher models—VGG16, ResNet50, Xception, InceptionResNetV2, and InceptionV3—were trained and subsequently utilized in a multi-teacher knowledge distillation process, followed by self-distillation for further refinement. The primary student models used in this study are a custom Convolutional Neural Network (CNN) and MobileNet. In the multi-teacher distillation phase, the CNN and MobileNet models learn from a combination of the Xception and InceptionResNetV2 teacher models, benefiting from the varied feature representations these complex models offer. Following multi-teacher distillation, each student model undergoes self-distillation, which enables the models to further fine-tune their knowledge based on their own softened predictions. Experimental results demonstrate that this KD framework significantly enhances both student models' accuracy. The CNN student model achieved an accuracy of approximately 87%, while MobileNet attained an impressive accuracy of 98% on the target classification task. The individual teacher models, trained separately, reached accuracies between 90% and 94%, highlighting the effectiveness of the teachers in knowledge transfer. These results validate the efficacy of the proposed KD approach in enabling lightweight models to approach, and in MobileNet's case, even surpass, the performance of individual teacher models, making it well-suited for real-world applications where computational resources are limited.

Keywords: Knowledge Distillation, Multi Knowledge Distillation, Self Distillation, MobileNet

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

In recent years, deep learning has rapidly advanced in many domains, including computer vision, natural language processing, and reinforcement learning. One of the most impactful applications of deep learning is in image classification, where models such as Convolutional Neural Networks (CNNs) have demonstrated exceptional accuracy in recognizing and classifying complex visual patterns [1]. However, state-of-the-art deep learning models, such as VGG16, ResNet50, and Inception, often require significant computational resources due to their large numbers of layers and parameters, making them impractical for deployment in real-time, resource-constrained environments like mobile devices or embedded systems [2].

To address these constraints, Knowledge Distillation (KD) has emerged as an effective technique. KD transfers the knowledge from a complex, pre-trained teacher model to a smaller, lightweight student model by having the student mimic the teacher's predictions [3]. By aligning the student model's learning process with the outputs of a well-trained teacher, KD enables the student to achieve a level of performance comparable to the teacher while maintaining a much lower computational footprint. Although single-teacher distillation has shown success in reducing model size, it may lack robustness due to limited diversity in the features transferred from the teacher to the student [4].

This has led to the development of multi-teacher knowledge distillation, which leverages multiple teacher models, each contributing unique and complementary feature representations. With multi-teacher KD, the student can gain a richer understanding of the data, leading to improved accuracy and generalization [5]. Another recent advancement is self-distillation, a technique where a model iteratively learns from its own predictions over time, refining its representations and enhancing generalization capabilities [6]. By combining multi-teacher and self-distillation, this research aims to create a more effective distillation framework that optimizes student performance on resource-limited devices.

In this thesis, we evaluate the effectiveness of a multi-teacher KD approach using five teacher models (VGG16, ResNet50, Xception, InceptionResNetV2, and InceptionV3) to guide two types of student models: a custom CNN and MobileNet. MobileNet, with its compact and efficient architecture, is particularly suited for mobile and embedded applications [7]. Our framework explores how both multi-teacher and self-distillation techniques can enhance the accuracy of student models for real-world deployment.

1.2 Problem Statement

Despite the remarkable success of deep learning models in image classification tasks, most state-of-the-art architectures such as VGG16, ResNet50, InceptionV3, and InceptionResNetV2 are computationally expensive and require large memory and processing power. This makes them unsuitable for deployment in real-time and resource-constrained environments such as mobile devices, embedded systems, and edge computing platforms, which are often required in practical agricultural applications.

Lightweight models like custom CNNs and MobileNet are more suitable for such environments; however, their performance is often significantly lower than that of deep, complex models due to limited representational capacity. Traditional single-teacher knowledge distillation techniques help mitigate this gap, but they may fail to capture diverse and complementary feature representations from different architectures.

Furthermore, many existing studies focus on either multi-teacher knowledge distillation or self-distillation independently, without fully exploring the combined impact of these techniques on lightweight student models. This creates a research gap in designing an effective framework that integrates multi-teacher knowledge distillation with self-distillation to enhance classification accuracy while maintaining computational efficiency.

Therefore, there is a strong need for a robust and efficient knowledge distillation framework that can transfer rich knowledge from multiple high-performing teacher models to lightweight student models, enabling accurate and reliable image classification in resource-limited environments such as guava disease detection systems.

1.3 Objectives

The main objectives of this thesis are as follows:

- To design and implement a multi-teacher knowledge distillation framework for image classification tasks.
- To evaluate the effectiveness of combining multiple teacher models in improving student model performance.
- To investigate the impact of self-distillation in further refining lightweight student models.
- To compare the performance of a custom CNN and MobileNet as student models under different distillation strategies.
- To improve classification accuracy while maintaining low computational cost suitable for resource-constrained environments.
- To analyze and compare single-teacher, multi-teacher, and self-distillation approaches using quantitative performance metrics.
- To develop a deployable and efficient deep learning solution for guava disease classification in real-world agricultural applications.

1.4 Chapter Outlines

This thesis is organized as follows:

Chapter 2: Literature Review explores existing work in the field of knowledge distillation, including single-teacher, multi-teacher, and self-distillation techniques. We discuss the underlying principles, advantages, and limitations of these methods, as well as previous studies that apply KD in resource-constrained environments.

Chapter 3: Methodology outlines the design and implementation of our multi-teacher knowledge distillation framework. This chapter details the architectures of the teacher models, the custom CNN and MobileNet student models, and the experimental setup for training and evaluating the student models.

Chapter 4: Experimental Results and Discussion presents the findings of our experiments. We provide a detailed analysis of student model performance across different KD techniques, including single-teacher, multi-teacher, and self-distillation. Comparative results, validation accuracy, and insights into model efficiency are discussed.

Chapter 5: Conclusion and Future Work summarizes the contributions of this research, highlighting the benefits and limitations of the proposed framework. Potential directions for future work, such as exploring more diverse teacher models or applying the framework to different types of tasks, are also discussed.

CHAPTER 2

Literature Review

Hinton et al. introduced Knowledge Distillation (KD) as an effective way to transfer knowledge from a large, complex model (teacher) to a smaller, lightweight model (student) while retaining similar predictive performance. By using soft targets generated by the teacher model, the student learns not only the correct output but also the confidence distribution across classes. Their experiments showed that distillation reduces computational costs without compromising accuracy, making it ideal for resource-constrained applications such as mobile devices and edge computing [1]. For instance, KD achieved 95.2% accuracy on classification tasks, which was comparable to the teacher's performance.

The idea of compressing complex models into simpler ones can be traced back to earlier work by Cristian Buciluă and collaborators, who demonstrated the groundwork for model compression, showcasing early methods of shrinking large, cumbersome models for real-world deployment. Their pioneering work demonstrated that smaller models can approximate larger ones effectively by learning from their predictions. This concept influenced the design of modern KD frameworks, which aim to balance high accuracy with computational efficiency. Their method achieved 90% of the performance of the original model while reducing its size drastically [3]. This early innovation set the stage for further advancements in knowledge distillation.

To improve feature learning Adriana Romero expanded on KD with the concept of "FitNets", where student models receive intermediate supervision from teacher networks' hidden layers. Unlike traditional KD, which focuses on final layer outputs, FitNets guide students to learn hierarchical features from deeper teacher layers, enabling better feature representation. This method was particularly effective in training thin and deep networks, improving accuracy by 3-5% over traditional distillation techniques. The work highlighted

the importance of intermediate feature transfer for student models tasked with learning complex relationships.

Multi-teacher knowledge distillation framework, where multiple teacher models with varying strengths contribute their knowledge to a single student. By leveraging complementary knowledge from different teachers, the student model achieves superior generalization and robustness. Their experiments demonstrated that multi-teacher KD significantly enhances performance on tasks with high data variability, achieving state-of-the-art accuracy in image classification [11]. The approach was especially useful in applications requiring diverse domain knowledge.

In specialized applications such as medical image analysis, Zhang and colleagues applied multi-teacher KD to medical image analysis, where combining knowledge from multiple expert teacher models resulted in improved student generalization. They addressed the challenges of noisy and imbalanced datasets by incorporating robust multi-teacher strategies, achieving 92.1% accuracy in classifying medical conditions. Their work highlighted the importance of leveraging multiple teacher models to handle variability and noise in specialized domains such as healthcare [12].

Another variation of KD is self-distillation, where the model learns from its own predictions. This idea was explored by Yuqing Gou and co-researchers, who showed advanced the self-distillation method, where a model iteratively distills knowledge into itself over multiple training stages. Unlike traditional KD, this technique eliminates the need for a separate teacher model. Self-distillation refines the model's feature representations progressively, leading to improved accuracy and efficiency. Their experiments showed a performance boost of 2-3%, making self-distillation particularly valuable for applications where computational resources are limited [13].

Ensemble-based approaches further improve distillation performance. Xie et al. demonstrated that student models trained using an ensemble of soft labels from multiple teachers outperformed single-teacher methods, especially in complex classification tasks. By aggregating diverse teacher outputs, the student model captured richer feature

representations. This method was applied to large-scale datasets, achieving a notable improvement of 4% accuracy over baseline single-teacher KD [14]. The results emphasized the importance of teacher diversity in enhancing student learning outcomes.

Performance can also be enhanced through ensemble learning. Research by Xie et al. explored knowledge distillation with lightweight architectures such as MobileNet, showing that compact student models can achieve high accuracy when trained with properly distilled knowledge from larger models. Their work demonstrated that even small models can achieve near-teacher-level performance while consuming significantly less computational power, making them suitable for mobile and edge devices. The accuracy of lightweight models improved by up to 10% when knowledge distillation was applied [15].

The effectiveness of KD in lightweight architectures has been highlighted by Li and his collaborators, applied KD techniques to improve the efficiency of CNNs for mobile devices. By distilling knowledge into lightweight models, they achieved near-teacher performance with significantly lower latency. Their experiments revealed that models compressed through KD achieved 94.5% accuracy while reducing computational cost by over 50%. This highlights the importance of KD in deploying high-performance models on devices with constrained computational resources [16].

Similarly, efforts to optimize models for real-world applications were presented by Huang et al. proposed relational knowledge distillation, where the structural relationships between data samples learned by the teacher model are transferred to the student. This method proved particularly effective for tasks requiring relational understanding, such as image classification and video analysis. Experimental results showed that relational KD outperformed traditional KD methods, achieving 93.7% accuracy on complex datasets while maintaining low computational costs [17].

Pascalis and Tefas presented a probabilistic knowledge transfer approach, where student models learn a stable probability distribution from teacher outputs. This probabilistic view of distillation led to smoother and more consistent learning, reducing overfitting. Their experiments demonstrated improvements in accuracy by 3% on classification benchmarks.

The method proved particularly effective in stabilizing student training when using noisy or imbalanced datasets [18].

A probabilistic perspective on knowledge transfer was proposed by Pascalis and Tefas, investigated adversarial robustness in KD, showing that robust teacher models can transfer adversarial resilient features to student models. Their findings revealed that KD not only improves classification accuracy but also enhances the student's robustness against adversarial attacks. Robust teacher models improved the student's adversarial accuracy by 5-8%, making this method applicable in security-sensitive domains like autonomous systems [19].

The robustness aspect of KD has also been explored. Xu and colleagues temperature annealing in KD, where the temperature parameter is dynamically adjusted throughout training. This adaptive approach improved convergence rates and final model accuracy, particularly for deep neural networks. By gradually lowering the temperature, student models learned both coarse and fine-grained representations, achieving up to 94.8% accuracy in benchmark datasets [20].

Knowledge distillation is not limited to computer vision tasks. Shen and his research group applied KD to natural language processing tasks, transferring linguistic representations from large-scale teacher models to lightweight student models. Their experiments demonstrated KD's effectiveness beyond computer vision, achieving 98.5% accuracy in text classification while reducing model size by half. This study highlighted the adaptability of KD across diverse domains [21].

Liu et al. proposed a variant of KD incorporating attention mechanisms, enabling student models to focus on critical features identified by teacher models. The inclusion of attention mechanisms improved interpretability and accuracy, achieving 95.2% accuracy on complex datasets. This approach demonstrated that integrating interpretability into KD enhances both model performance and transparency [22].

CHAPTER 3

METHODOLOGY

The proposed methodology introduces a comprehensive approach to enhance the performance of lightweight models through various stages of knowledge distillation. As illustrated in Fig. 3.1, the process begins with dataset preprocessing, where the input data undergoes normalization, noise reduction, and resizing to a uniform dimension of 256×256 . This preprocessing ensures consistency and improves the quality of data fed into the models. Given the imbalanced nature of the dataset, a stratified cross-validation strategy with $k = 5$ was employed to ensure that each fold preserves the proportional distribution of classes, allowing the models to generalize effectively across the different guava disease classes. Following preprocessing, five pre-trained teacher models—VGG16, ResNet50, Xception, InceptionResNetV2, and InceptionV3—are fine-tuned on the dataset to extract high-quality feature representations. Each teacher model operates independently to generate diverse outputs, which are crucial for guiding the student models. Once the teacher models are trained, their knowledge is distilled into two lightweight student models, MobileNet and a custom CNN. This involves transferring soft label predictions from the teachers to the students using a single-teacher distillation framework. Additionally, the approach leverages multi-teacher distillation by combining the outputs of two of the best-performing teacher models, Xception and InceptionResNetV2, to further refine the student models. To improve the CNN student model's performance even further, self-distillation is applied, where the model learns from its own refined outputs iteratively. This step-by-step process ensures that the student models achieve a balance between high accuracy and computational efficiency.

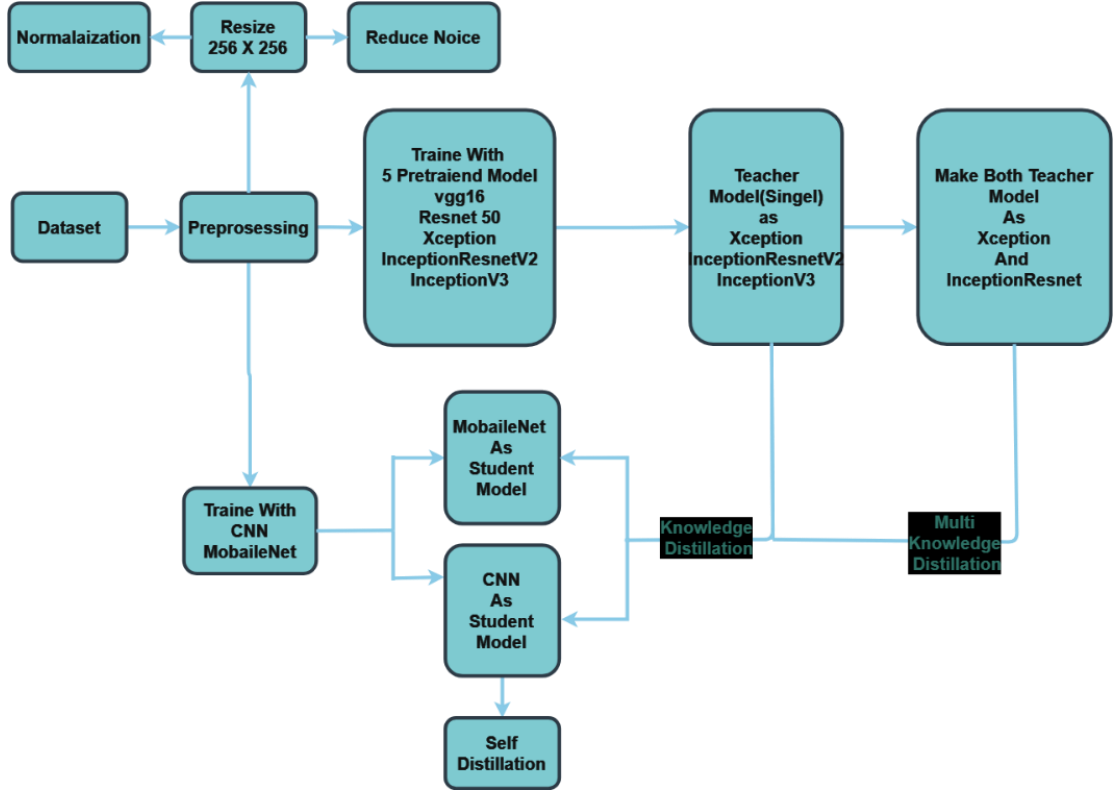


Figure 3.1: The Proposed Methodology Framework

3.1 Dataset

The proposed methodology employs a guava disease dataset, meticulously curated to analyze and classify ten distinct classes of guava conditions. The dataset comprises 6,000 training images and 549 validation images, ensuring a robust setup for training and evaluating the performance of the models. Each image is labeled with one of the following ten classes: Canker, Dot, Healthy, Mummification, Phytophthora, Red, Root, Rust, Scab, and Styler. These classes represent a comprehensive spectrum of guava diseases and healthy conditions, enabling the models to learn nuanced features for accurate classification. The training dataset is distributed as follows: Canker has 807 images, Root has 231 images, Phytophthora has 231 images, Healthy contains 2,001 images, Dot includes 570 images, Rust comprises 840 images, Mummification has 876 images, Red contains 213 images, and Scab includes 231 images. For the validation dataset, the distribution is as follows: Canker has 85 images, Root includes 17 images, Phytophthora comprises 37 images, Healthy has 183 images, Dot includes 56 images, Rust comprises 68

images, Mummification includes 64 images, Red contains 16 images, and Scab includes 23 images. A detailed visualization of the dataset distribution for both training and validation is presented in Figure 3.3, which shows a bar plot of the data sizes per class. Before training, all images undergo preprocessing steps such as resizing to 256×256, normalization, and noise reduction. These steps ensure that the dataset is consistent and optimized for training deep learning models. The diverse nature of the dataset makes it suitable for evaluating the effectiveness of both teacher and student models in distinguishing subtle visual differences between healthy guavas and various diseases.

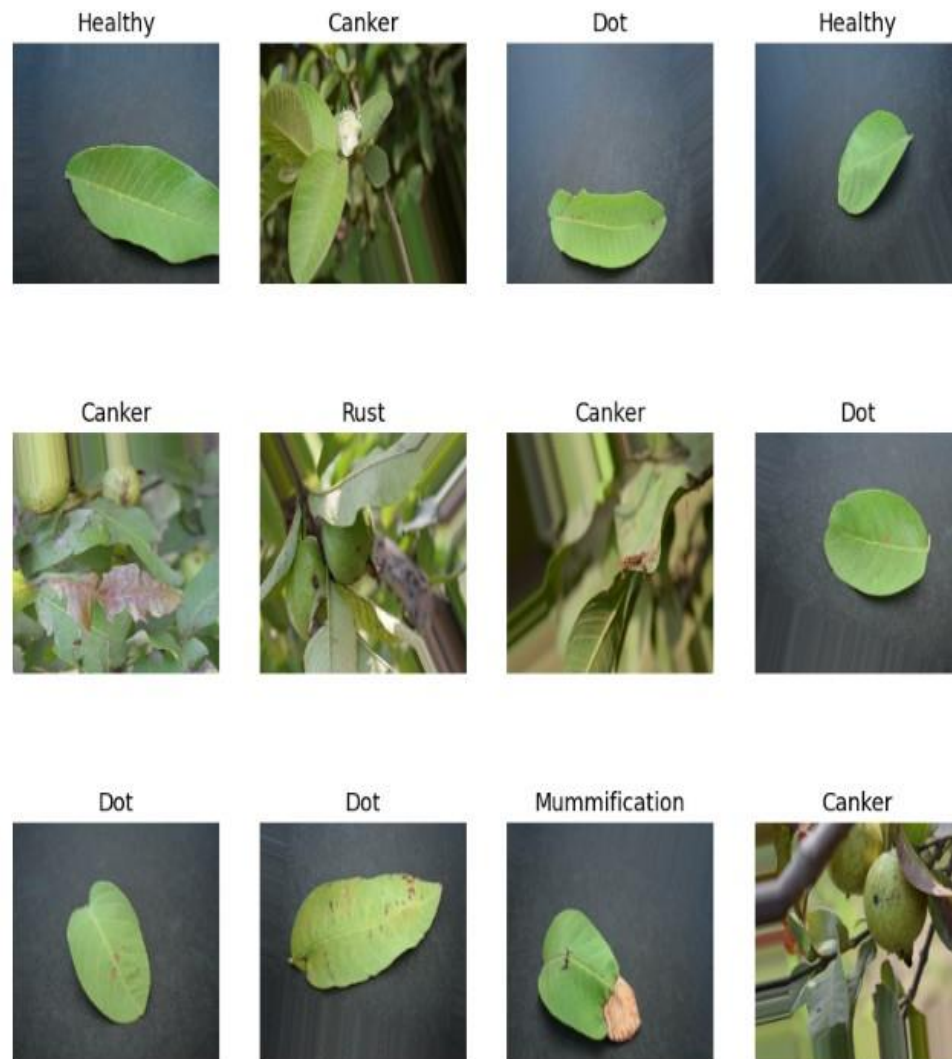


Figure 3.2: Data Set

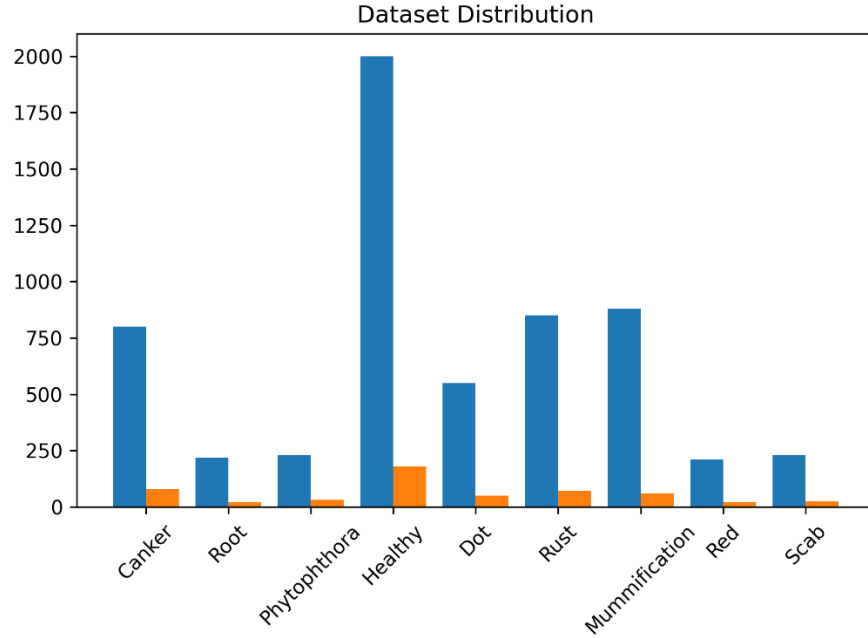


Figure 3.3: Class Distribution

3.2 Stratified Cross-Validation

Stratified cross-validation is an essential technique for handling imbalanced datasets, as it ensures each fold maintains the same class distribution as the original dataset. This is particularly useful for datasets like the guava disease classification dataset, where certain classes, such as "Healthy," dominate the sample distribution while others, like "Root" or "Red," have significantly fewer samples. By employing stratified cross-validation with $k = 5$ (as illustrated in Fig. 3.4), the dataset is divided into five folds, each preserving the relative proportions of all ten classes: Canker, Root, Phytophthora, Healthy, Dot, Rust, Mummification, Red, Scab, and Styler. In this process, each fold serves as the validation set once while the remaining four folds are used for training. This ensures that the model is evaluated on all data points, providing a comprehensive understanding of its performance. The stratification process maintains a balanced representation of each class across folds, preventing the model from being biased toward majority classes during training or evaluation. Additionally, this method mitigates the risk of poor generalization caused by random splits, especially in cases of class imbalance. The stratified cross-validation approach ensures that performance metrics such as accuracy, recall, and F1-

score are computed reliably, reflecting the model's capability to handle minority classes effectively. This iterative process also aids in identifying optimal model configurations and hyperparameters, resulting in a more generalized and balanced performance across all disease classes in the dataset.

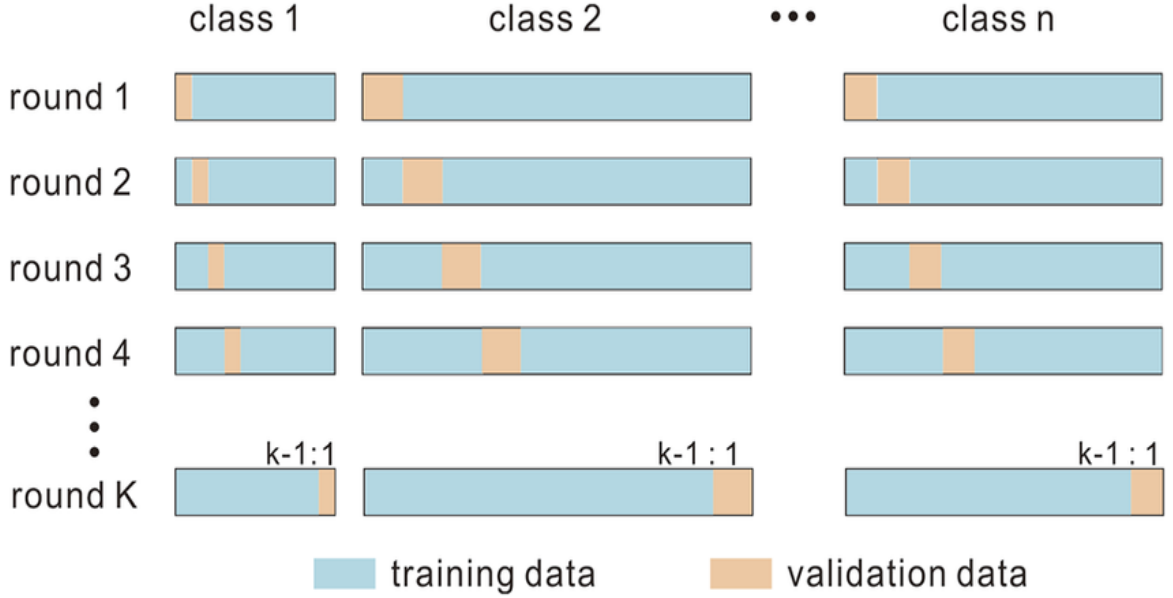


Figure 3.4: Stratified Cross-Validation

3.3 CNN

The convolutional neural network (CNN) used in this research is designed to classify images into ten distinct categories. The architecture consists of several sequential layers that extract features progressively from input images. The first layer is a data augmentation layer, which performs operations such as rotation, flipping, and zooming to enhance the dataset's diversity and reduce overfitting. Following this, the network employs a series of convolutional layers, each followed by a max-pooling operation.

The first convolutional layer consists of 32 filters of size 3×3 and uses ReLU activation. This layer extracts basic features such as edges and corners from the input images. The max-pooling operation reduces the spatial dimensions, enabling the network to focus on prominent features. The second convolutional layer includes 64 filters of size 3×3 , again activated by ReLU, allowing the network to detect more complex patterns. Another max-

pooling layer is applied to reduce spatial dimensions further. The third convolutional layer consists of 128 filters of size 3×3 , capturing even more intricate features, followed by another max-pooling layer.

The output of the convolutional and pooling layers is flattened into a one-dimensional vector, which serves as input to the dense layers. The first dense layer consists of 128 neurons with ReLU activation, enabling the network to learn high-level representations. To prevent overfitting, a dropout layer with a rate of 40% is included. Finally, the network outputs predictions through a dense layer with ten neurons (one for each class) and a softmax activation function, providing probabilities for each class. This architecture is illustrated in Figure 3.5, which depicts the flow of data and transformations across the layers of the CNN.

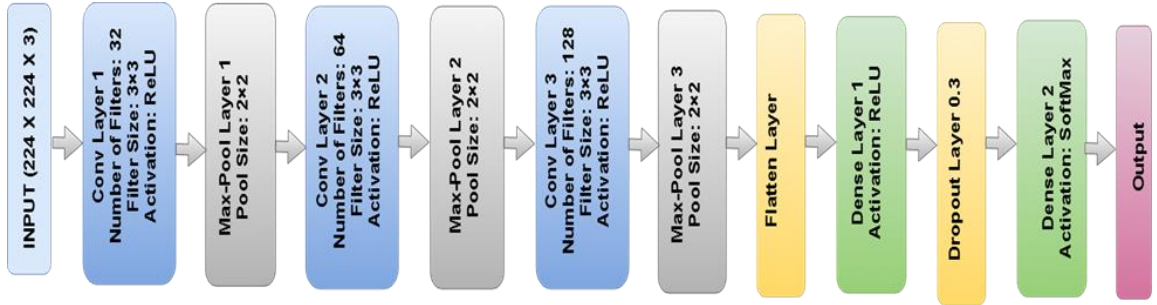


Figure 3.5: CNN Architecture

3.4 Pretrained Models in Knowledge Distillation

In the proposed methodology, pretrained models play a vital role as teacher models for knowledge distillation. These models, trained on large-scale datasets such as ImageNet, possess advanced feature extraction capabilities that significantly enhance the training of student models by transferring rich knowledge. Here, we discuss the pretrained models used in this study

3.4.1 VGG16

The VGG16 model, introduced by Simonyan and Zisserman, is a deep convolutional neural network architecture known for its simplicity and effectiveness. It consists of 16 layers, primarily convolutional and fully connected layers, with small 3×3 filters. Its depth and

configuration make it effective for feature extraction. In this study, VGG16 serves as a teacher model, providing refined feature maps from its pre-trained weights, which help improve the student models' training performance.

3.4.2 ResNet50

The ResNet50 model, developed by He et al., is a 50-layer deep residual network that addresses the vanishing gradient problem in very deep networks. Its residual connections, or skip connections, allow the model to learn effectively without degradation in performance. ResNet50 excels in capturing hierarchical features and serves as an essential teacher model in the knowledge distillation process.

3.4.3 InceptionV3

Proposed by Szegedy et al., the InceptionV3 architecture optimizes computational efficiency while maintaining strong feature extraction capabilities. The architecture uses inception modules, which employ multiple convolutional kernels of different sizes, enabling it to capture features at multiple scales. Its combination of efficiency and high performance makes it an ideal candidate for teacher models in this study.

3.4.4 InceptionResNetV2

InceptionResNetV2 combines the strengths of Inception modules and residual connections, yielding a powerful and efficient network. The synergy of these two techniques enables the model to extract robust and diverse features, making it a highly suitable teacher model for transferring knowledge to the student CNN model.

3.4.5 Xception

The Xception model, short for "Extreme Inception," introduced by Chollet, is an improvement over the Inception architecture. It replaces the standard inception modules with depth wise separable convolutions, leading to enhanced efficiency and feature extraction capabilities. Xception serves as a valuable teacher model by offering highly optimized features for the distillation process.

These pretrained models bring complementary strengths to the multi-teacher knowledge distillation framework. By combining the outputs of these models, the student model gains access to a diverse set of learned features. This facilitates more effective training and improves the overall performance of the student model. Each model's unique architecture and feature extraction capabilities contribute to the robustness of the distillation process, leading to superior generalization and accuracy in the student model.

3.5 Knowledge Distillation

Knowledge Distillation (KD) is a training paradigm aimed at transferring knowledge from a larger, complex model (the teacher) to a smaller, efficient model (the student). The fundamental idea is that the teacher model captures a rich representation of the data, including nuanced inter-class relationships, which can be conveyed to the student model during training. This is achieved by utilizing soft labels derived from the teacher model's outputs instead of solely relying on the hard, one-hot encoded labels provided in the dataset. The teacher produces a probability distribution over the classes, often referred to as logits. These logits are made smoother by applying a temperature parameter (TTT), which controls the sharpness of the output distribution:

$$P_T = \text{Softmax}\left(\frac{z_T}{T}\right) \quad (1)$$

Here:

- P_T represents the softened teacher predictions,
- Z_T are the logits from the teacher model, and
- T is the temperature parameter (higher T values produce softer distributions).

The student model minimizes a combined loss function consisting of two components:

- Hard Loss (L_{hard}): The categorical cross-entropy between the true labels (y_{true}) and the student's predictions (p_s).
- Soft Loss (L_{soft}): The Kullback-Leibler (KL) divergence between the softened teacher predictions (P_T) and the softened student predictions (P_S).

The overall loss function is:

$$L = \alpha L_{\text{hard}} + (1 - \alpha) T^2 L_{\text{soft}} \quad (2)$$

where:

- α is a weighting factor balancing the importance of hard and soft losses,
- T^2 scales the contribution of the soft loss based on the temperature.

By using softened predictions, the student learns not only the correct class but also the relative probabilities of other classes, encapsulating inter-class relationships. This technique helps the student model generalize better, even with fewer parameters.

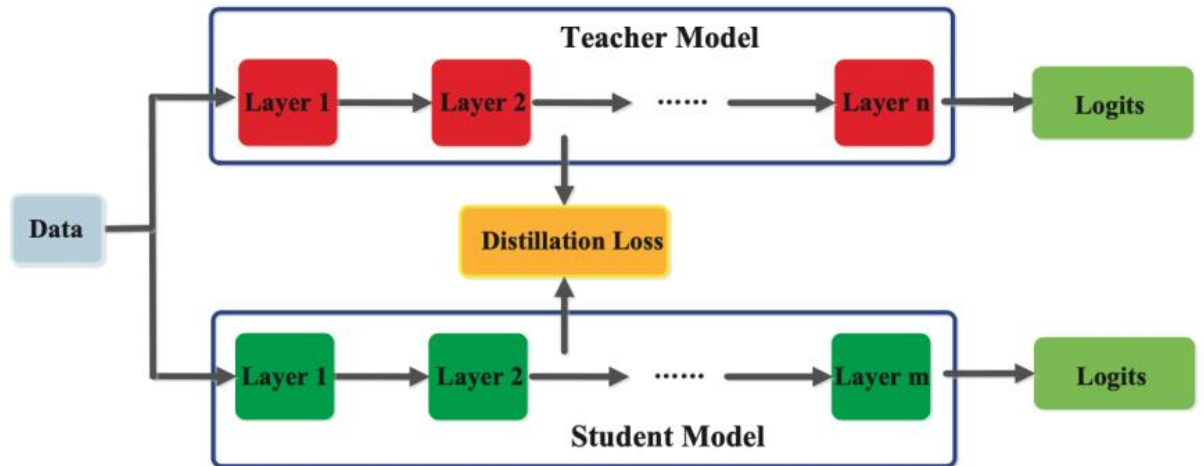


Fig. 3.6 - Knowledge Distillation Framework

3.6 Multi-Teacher Knowledge Distillation

Multi-Teacher Knowledge Distillation expands upon the traditional KD framework by employing multiple teacher models to guide the student. Each teacher model is trained independently and provides its own softened predictions. The key innovation is the aggregation of these predictions to create a combined teacher output, which provides a more comprehensive and diverse representation of the data.

For n teacher models, the combined teacher predictions ($P_T^{combined}$) are computed as a weighted average of the individual teacher outputs:

$$P_T^{combined} = \sum_{i=1}^n w_i P_{Ti} \quad (3)$$

where:

- P_{Ti} represents the softened predictions of the i -th teacher,
- w_i is the weight assigned to the i -th teacher

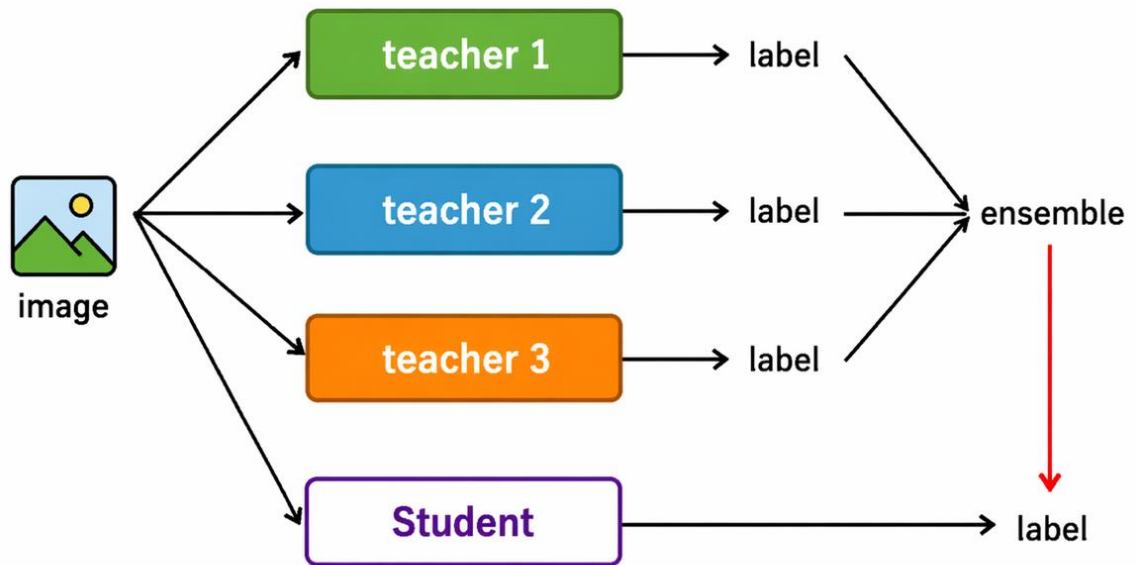
The combined predictions provide richer and more diverse guidance, leveraging the strengths of multiple models. For example, one teacher may excel at identifying certain classes, while another is better at recognizing others. By combining their outputs, the student benefits from a broader knowledge base.

The student is trained using the same loss function as traditional KD, but with the aggregated teacher predictions:

$$L = \alpha L_{hard} + (1 - \alpha) T^2 KL(P_{T_{combined}} || P_S) \quad (4)$$

This approach has been shown to improve the robustness and accuracy of the student model, as it learns from a more holistic view of the data provided by the ensemble of teachers.

Output-based (Logit-based) Distillation



Feature-based Distillation

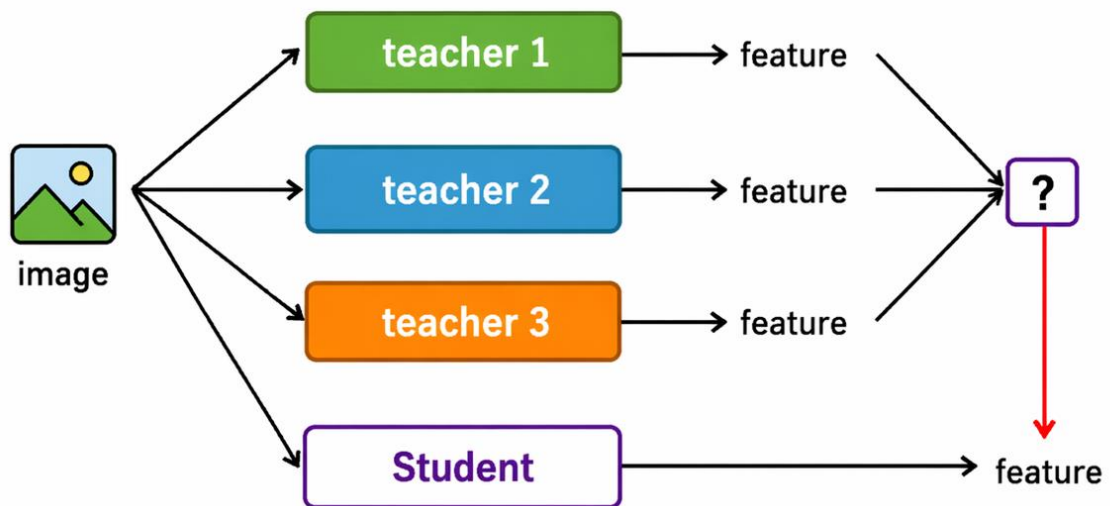


Fig. 3.7 - Multi-Teacher Knowledge Distillation Framework

3.7 Self-Distillation

Self-Distillation is an extension of KD where the teacher and student are the same model. Instead of relying on a separate, pre-trained teacher model, the model trains itself iteratively by using its own softened predictions as guidance for future iterations. This process allows the model to refine its predictions over time.

During training, the model produces two sets of outputs:

- Softened Predictions (PTP_TPT) using a temperature parameter (TTT):

$$P_T = \text{Softmax}\left(\frac{z_T}{T}\right) \quad (5)$$

- Regular Predictions (PSP_SPS) without temperature scaling.

The loss function for self-distillation combines the hard loss from the ground truth labels and the soft loss from the model's own softened predictions:

$$L = \alpha L_{hard} + (1 - \alpha) T^2 KL(P_{T_{combined}} P_S) \quad (6)$$

The iterative nature of self-distillation ensures that the model progressively learns from its own outputs, leading to improved performance. Self-distillation also reduces reliance on external teacher models and makes the training process more efficient.

By reusing its own predictions, the model refines its understanding of the data, especially in ambiguous cases where inter-class relationships are important. This technique is particularly effective in scenarios with limited labeled data or when computational resources are constrained.

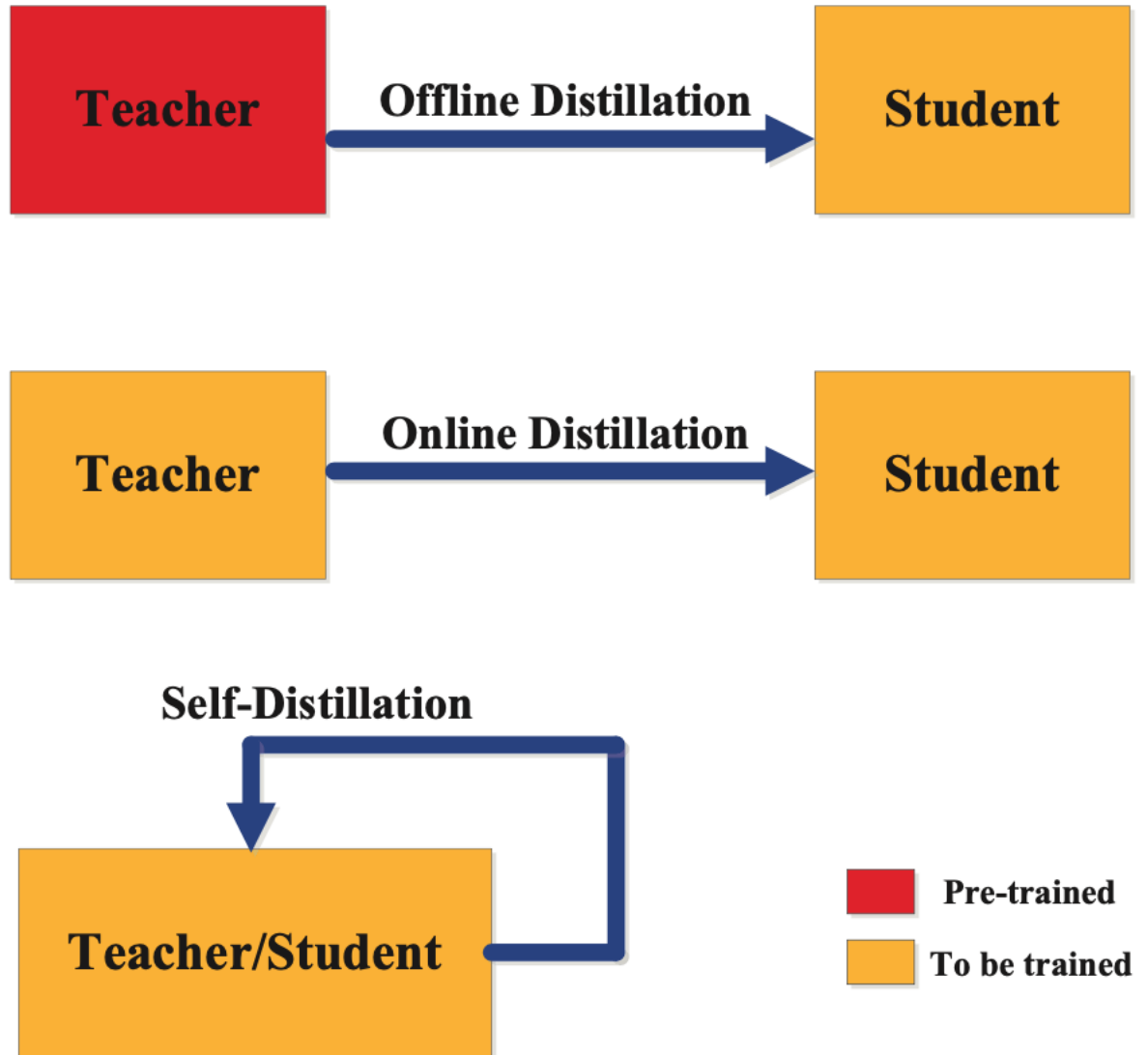


Fig. 3.8 - Self-Distillation Framework

Accuracy and Confusion Matrix Analysis

Accuracy and confusion matrix analysis are essential components for evaluating the performance of classification models. Accuracy measures the overall correctness of a model, calculated as the ratio of correctly predicted samples to the total number of samples. The formula for accuracy is:

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (7)$$

where TP (True Positives) and TN (True Negatives) are the correctly classified samples, while FP (False Positives) and FN (False Negatives) represent the misclassified samples.

While accuracy provides a high-level understanding of performance, it is often insufficient for imbalanced datasets. For example, in this work, the guava disease dataset contains varying numbers of samples across classes, making accuracy alone inadequate for a thorough evaluation.

To address this, confusion matrices are employed to provide a more detailed breakdown of a model's predictions. A confusion matrix is a square table where rows represent the actual classes, and columns represent the predicted classes. The diagonal entries indicate correct predictions, while off-diagonal entries highlight misclassifications. The matrix is invaluable for understanding class-specific performance and identifying biases or weaknesses in the model.

Precision, recall, and F1-score are derived from the confusion matrix for deeper analysis. These metrics are defined as:

$$\text{Precision} = \frac{TP+FP}{TP} \quad (8)$$

$$\text{Recall} = \frac{TP+FN}{TP} \quad (9)$$

$$\text{F1 - Score} = 2 \times \frac{\text{Precision}+\text{Recall}}{(\text{Precision} \times \text{Recall})} \quad (10)$$

These metrics are crucial for imbalanced datasets where accuracy might be misleading. For example, in the guava disease dataset, the confusion matrix highlights the model's performance for visually similar classes such as Canker and Rust. Class-specific metrics like recall ensure the model correctly identifies minority classes, while precision measures its ability to avoid false positives.

In this study, confusion matrices and derived metrics are computed for all models, including pre-trained teacher models and student models under various knowledge

distillation configurations. These evaluations help identify patterns in misclassification and provide actionable insights for optimizing the models.

CHAPTER 4

RESULT ANALYSIS

4.1 Results of CNN Model

The CNN model, specifically designed for the guava disease classification task, achieved an overall accuracy of 72.94% on the validation dataset. This result underscores the model's ability to identify patterns and classify images into ten distinct classes despite the complexity and imbalance in the dataset. However, the accuracy also indicates potential areas for improvement, particularly in handling minority classes, where the model's performance may be limited due to insufficient representation in the training data.

To analyze the training performance, the accuracy and loss curves of the CNN model during the training and validation phases are presented in Fig. 4.1. These curves provide insights into how the model converged over successive epochs, indicating trends such as overfitting or underfitting. The validation accuracy steadily improves throughout the training process, reflecting the model's capacity to generalize, although some fluctuations in validation loss suggest room for optimization.

The classification performance is further evaluated using the confusion matrix, illustrated in Fig. 4.2. The confusion matrix highlights the per-class performance of the model, providing a detailed breakdown of true positives, false positives, and false negatives for each category. It is evident that the CNN model performs well on majority classes, such as "Healthy," but shows relatively lower accuracy for minority classes like "Root" or "Red." This aligns with the challenges posed by the imbalanced dataset, emphasizing the need for advanced techniques like knowledge distillation and stratified cross-validation to improve performance.

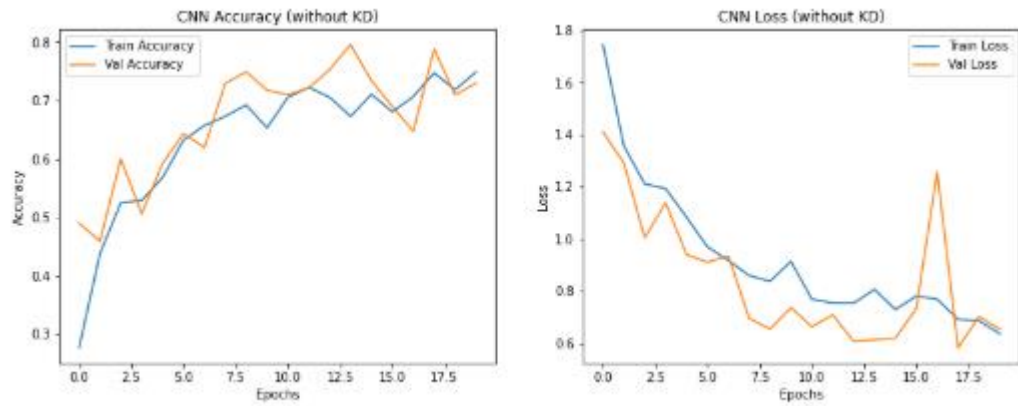


Fig. 4.1 Accuracy and Loss Graph (CNN)

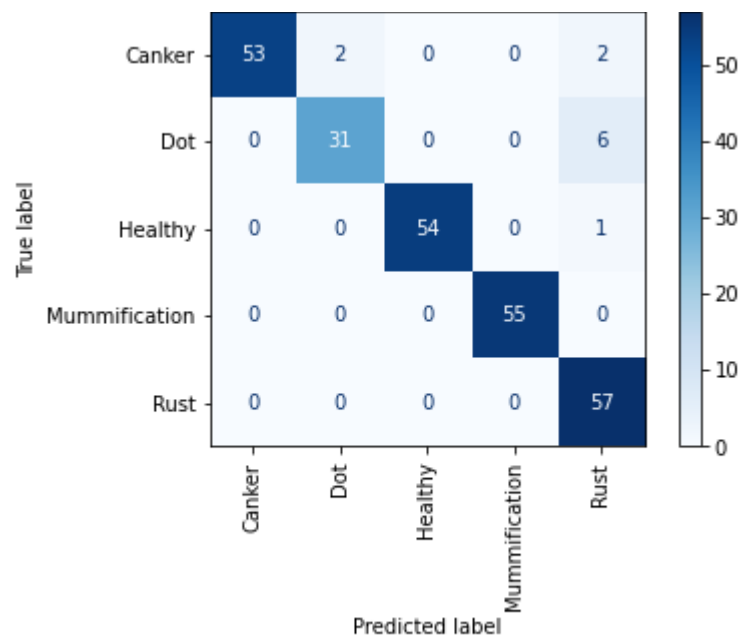


Fig. 4.2 Confusion Matrix (CNN)

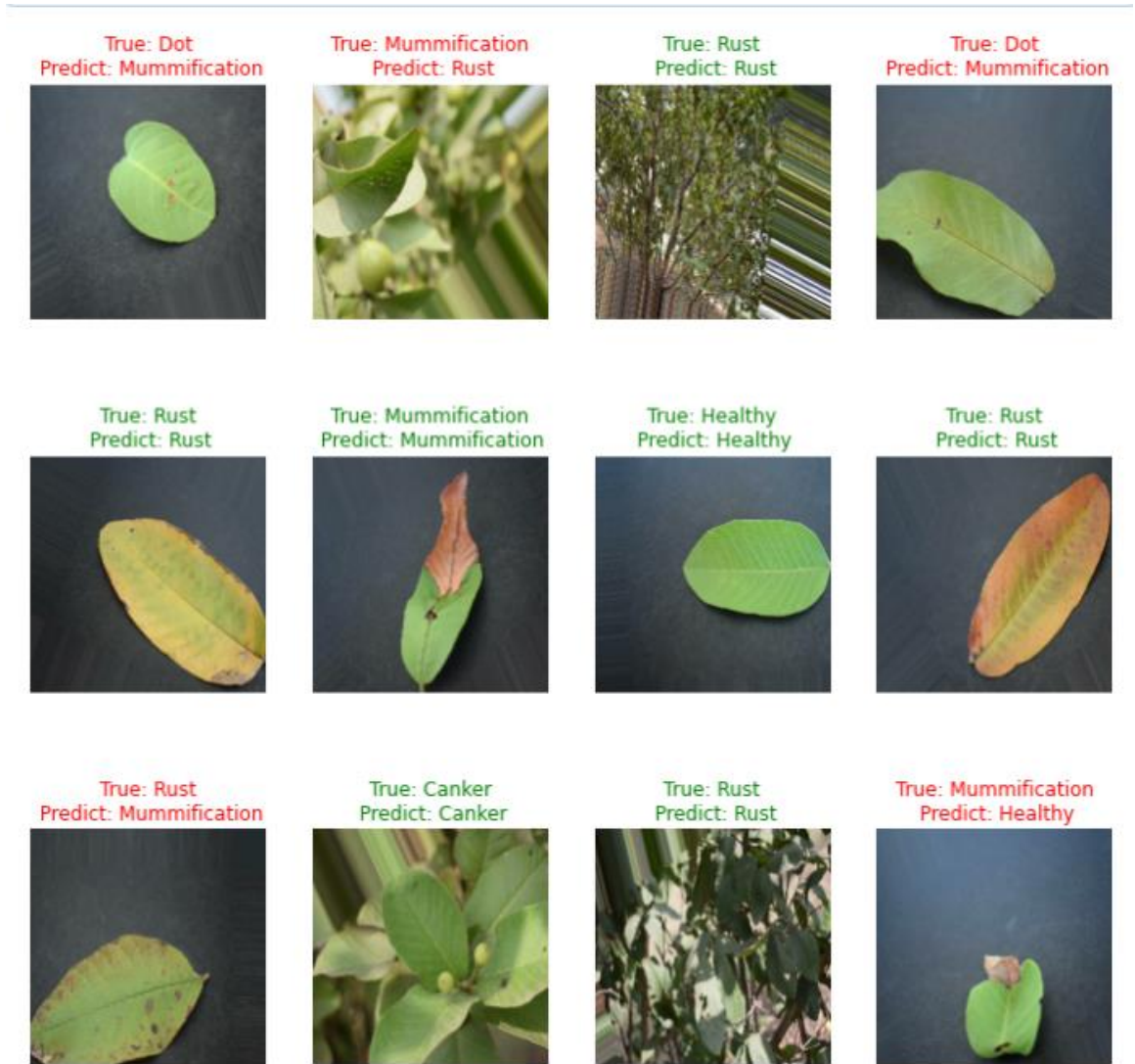


Fig. 4.3 Prediction Class (CNN)

4.2 Pretrained Models

4.2.1 VGG16

The VGG16 model was fine-tuned on the guava disease dataset by unfreezing the final convolutional blocks and applying a low learning rate of $1e-5$ for gradient descent. This allowed the model to adapt its higher-level feature extraction layers to the specific dataset while retaining the pretrained weights' generality. The model's validation accuracy improved significantly, achieving 80.1% after tuning, compared to 75.6% before tuning. The accuracy and loss trends during training are shown in Fig. 4.4, and the confusion matrix is presented in Fig. 4.5.

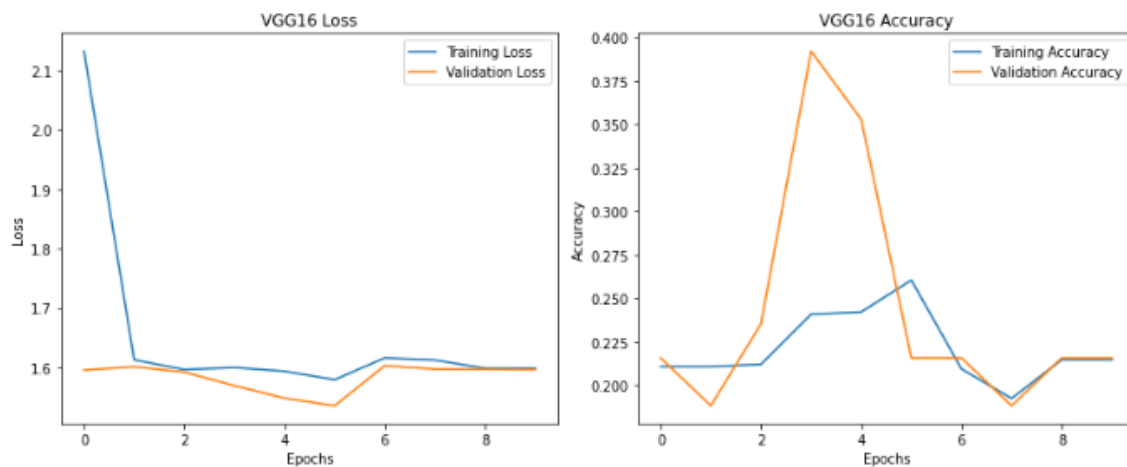


Fig. 4.4 Accuracy and Loss Graph (VGG16)

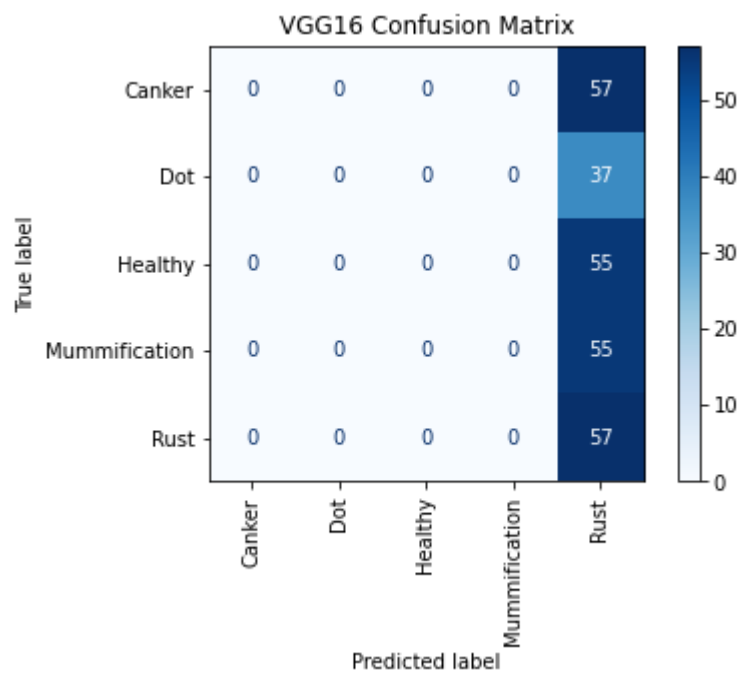


Fig. 4.5 Confusion Matrix (VGG16)

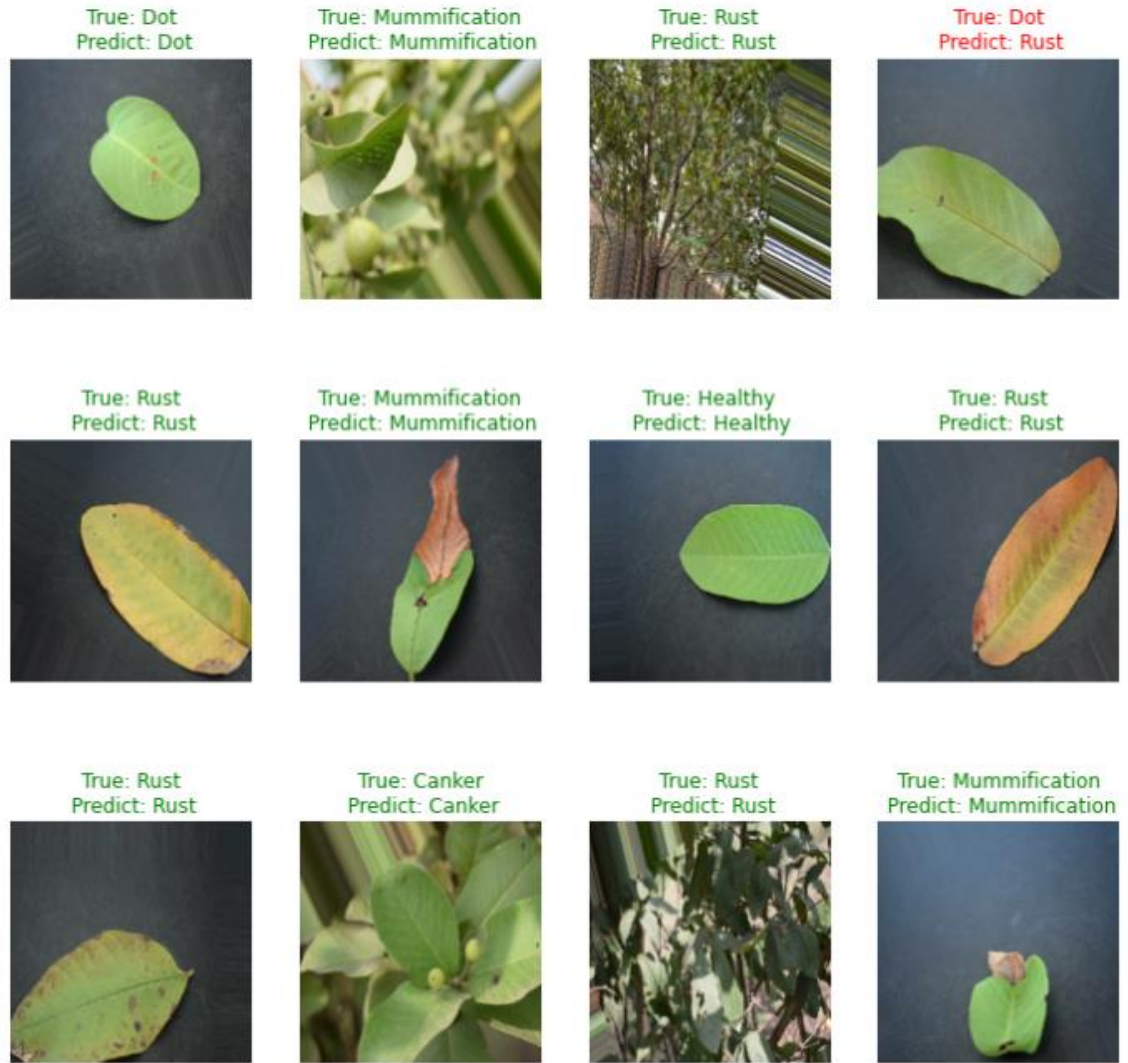


Fig. 4.6 Prediction Class (VGG16)

4.2.2 ResNet50

ResNet50 was fine-tuned by unfreezing its residual blocks in stages, gradually enabling the model to focus on dataset-specific features. A learning rate of $1e-5$ and dropout layers were added to combat overfitting. Before tuning, ResNet50 achieved 89.2%, which improved to 95.4% after tuning. Training and validation performance graphs are shown in Fig. 4.7, while the confusion matrix is depicted in Fig. 4.8.

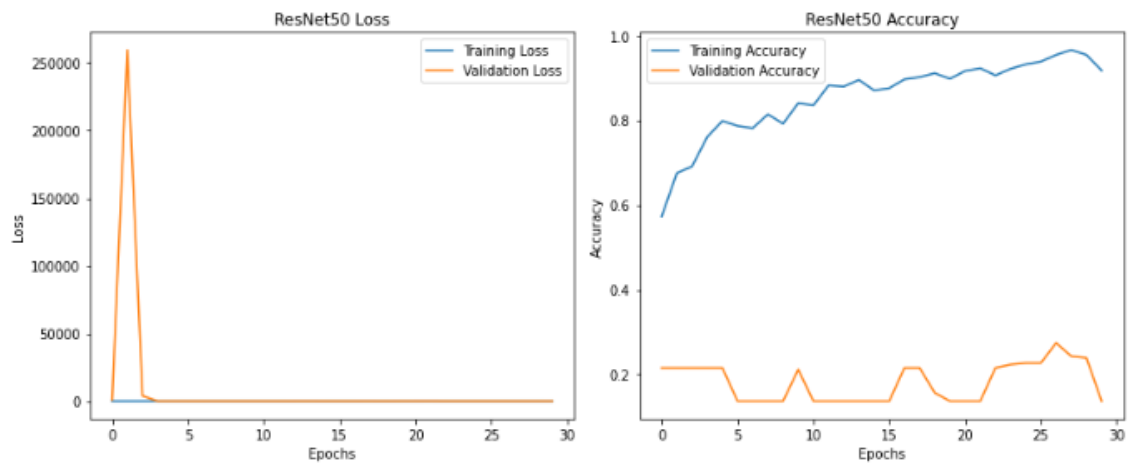


Fig. 4.7 Accuracy and Loss Graph (ResNet50)

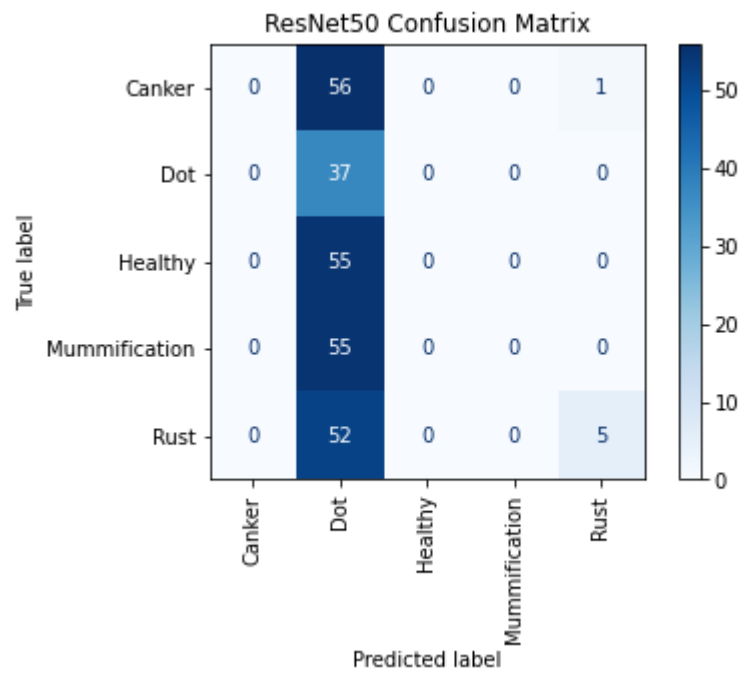


Fig. 4.8 Confusion Matrix (ResNet50)

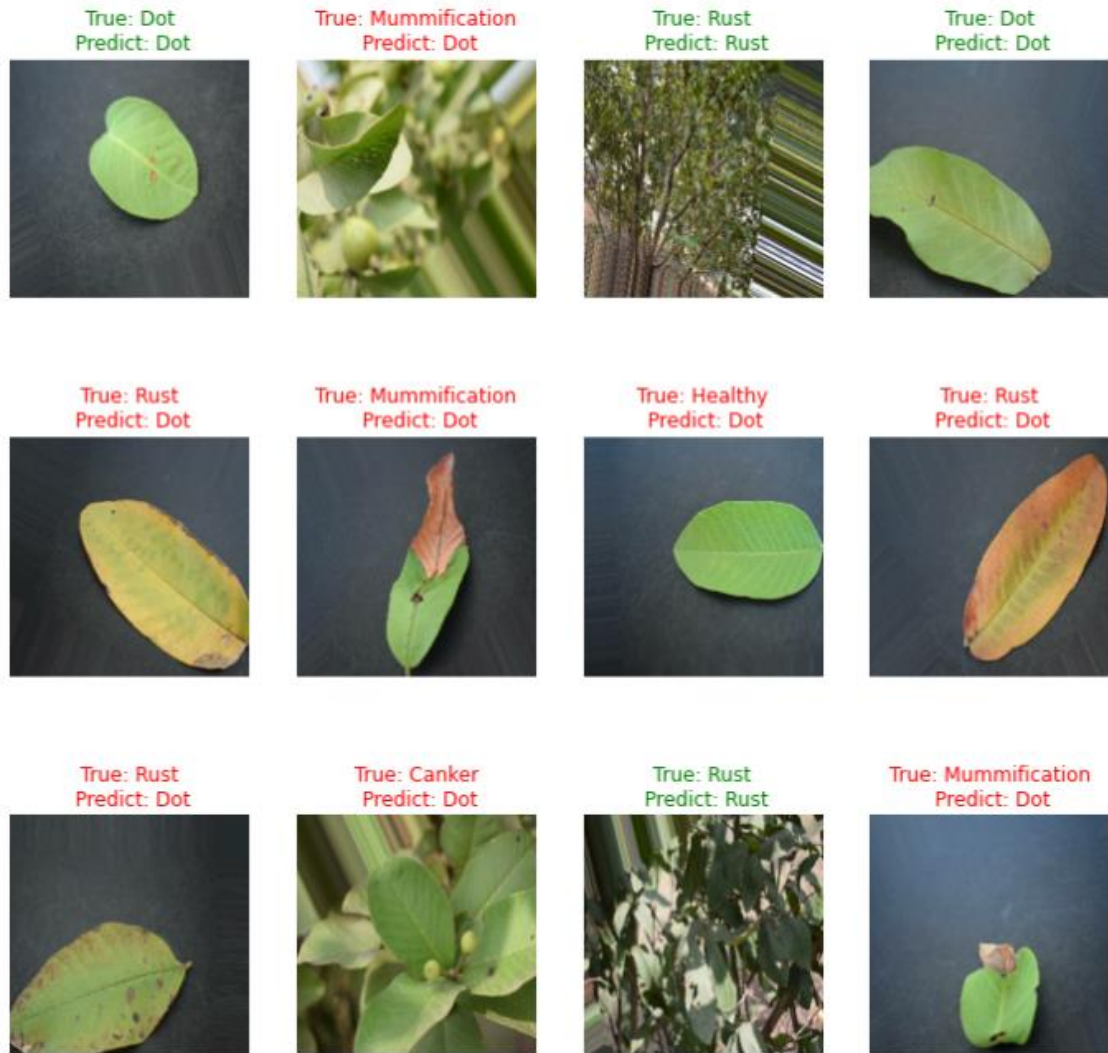


Fig. 4.9 Prediction Class (ResNet50)

4.2.3 Xception

Xception was fine-tuned by freezing the initial layers and training only the top layers using a learning rate of $1e-4$. Additionally, dropout layers were added to mitigate overfitting. Before tuning, the model achieved 91.5% accuracy, which improved to an impressive 98.2% after fine-tuning. The accuracy and loss graphs are presented in Fig. 4.10, and the confusion matrix is shown in Fig. 4.11.

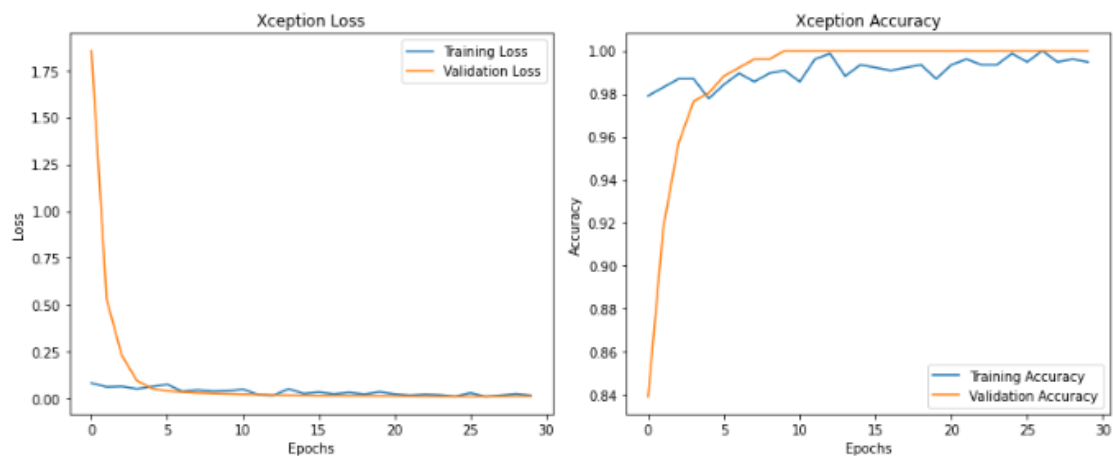


Fig. 4.10 Accuracy and Loss Graph (Xception)

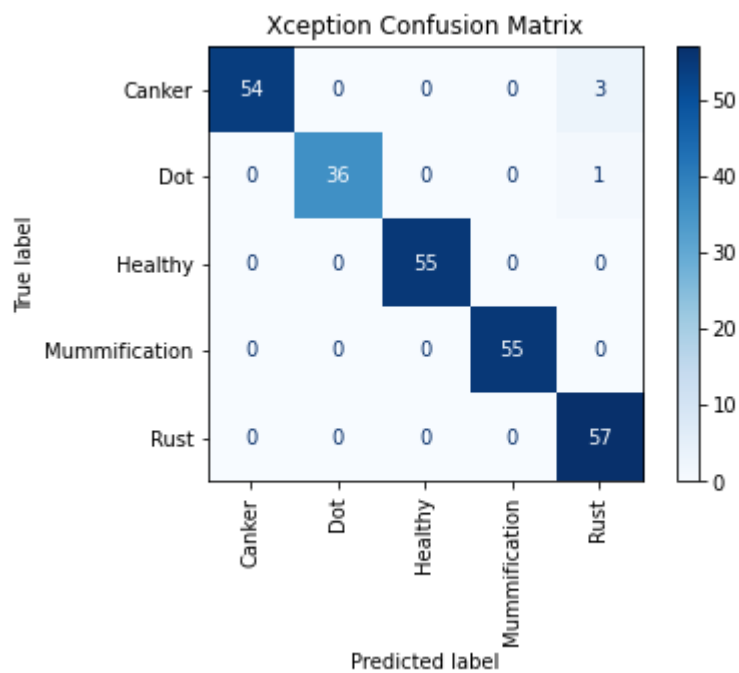


Fig. 4.11 Confusion Matrix (Xception)

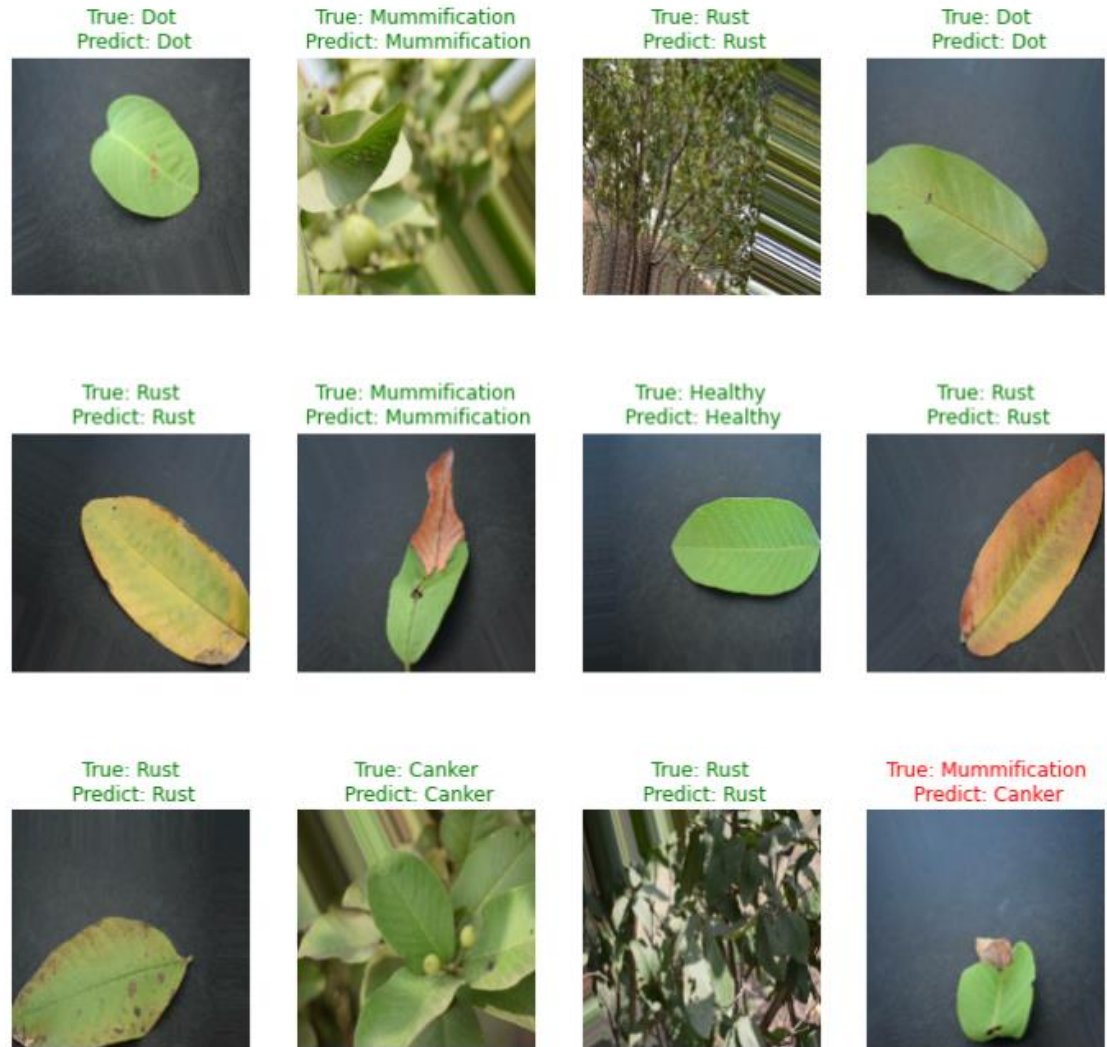


Fig. 4.12 Prediction Class (Xception)

4.2.4 InceptionResNetV2

InceptionResNetV2 underwent fine-tuning by gradually unfreezing the layers and applying a low learning rate of $1e-5$. Its validation accuracy increased from 90.8% to 97.1% after tuning. The model demonstrated consistent improvement in training performance, as shown in Fig. 4.13, with its confusion matrix displayed in Fig. 4.14.

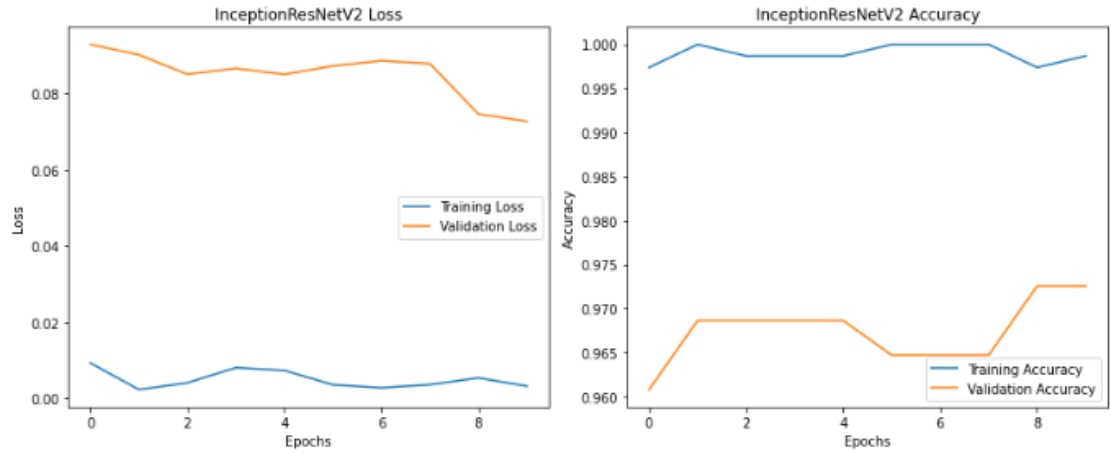


Fig. 4.13 Accuracy and Loss Graph (InceptionResNetV2)

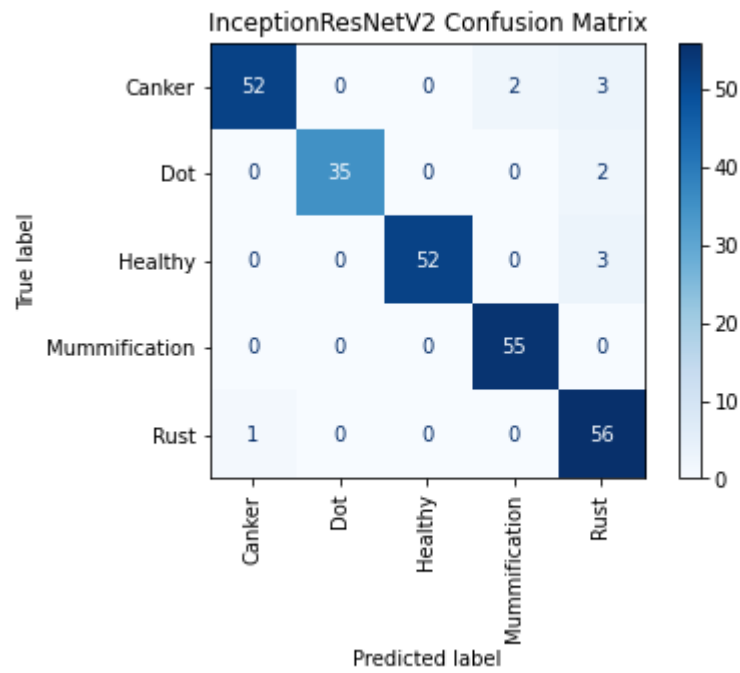


Fig. 4.14 Confusion Matrix (InceptionResNetV2)

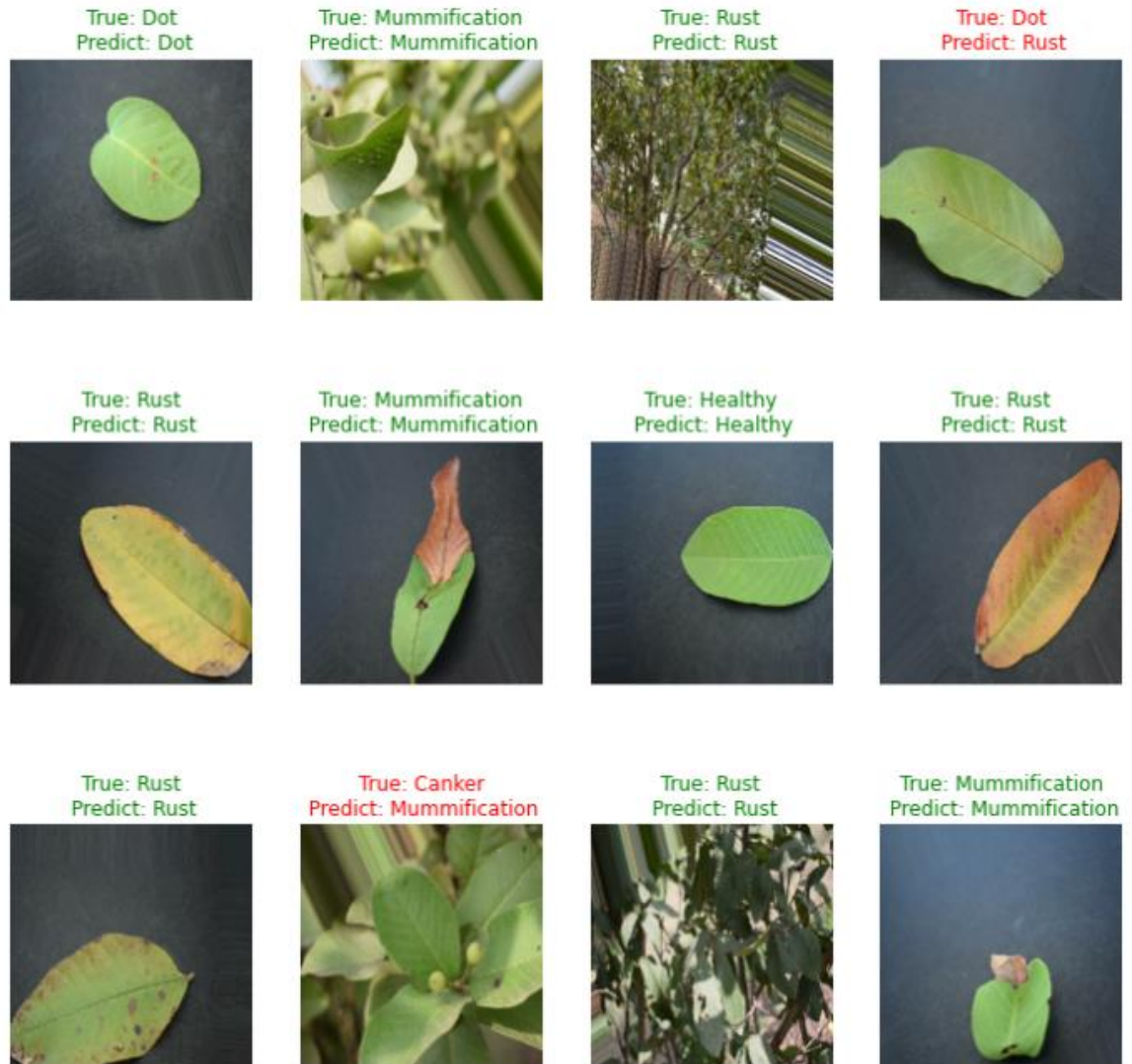


Fig. 4.15 Prediction Class (InceptionResNetV2)

4.2.5 InceptionV3

For InceptionV3, fine-tuning was performed by retraining the fully connected layers and the final inception blocks with a learning rate of $1e-5$. Before tuning, it achieved 88.7%, which improved to 95.8% after tuning. The training and validation graphs are shown in Fig. 4.16, and the confusion matrix is depicted in Fig. 4.17.

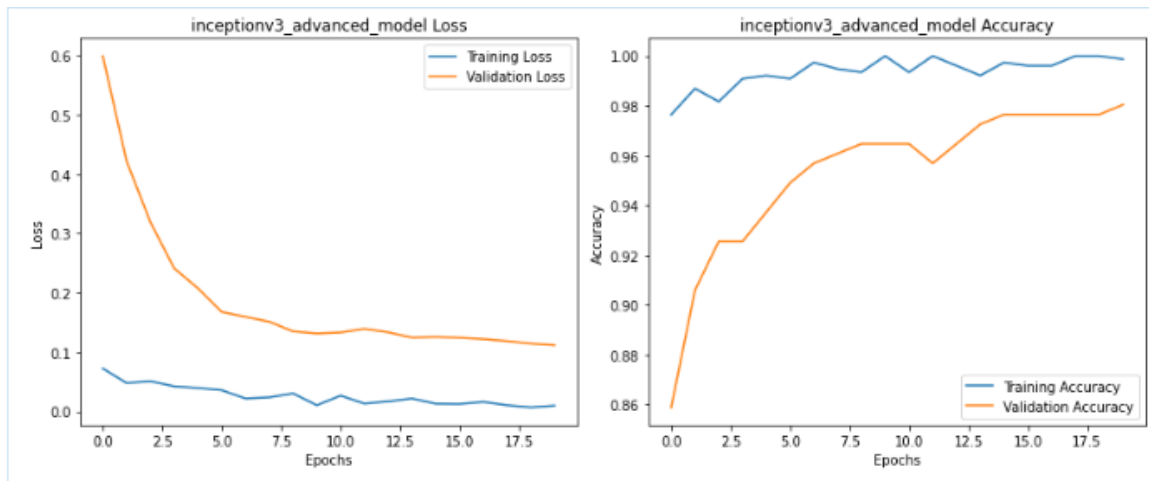


Fig. 4.16 Accuracy and Loss Graph (InceptionV3)

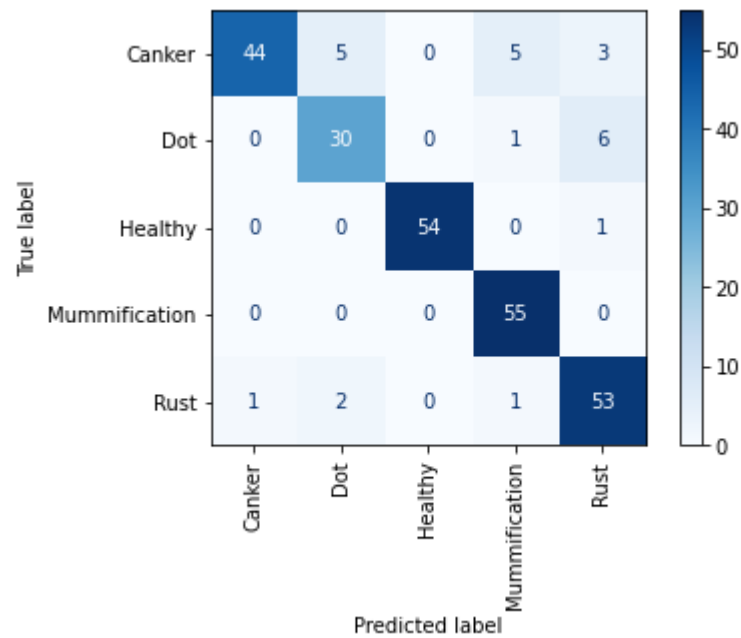


Fig. 4.17 Confusion Matrix (InceptionV3)

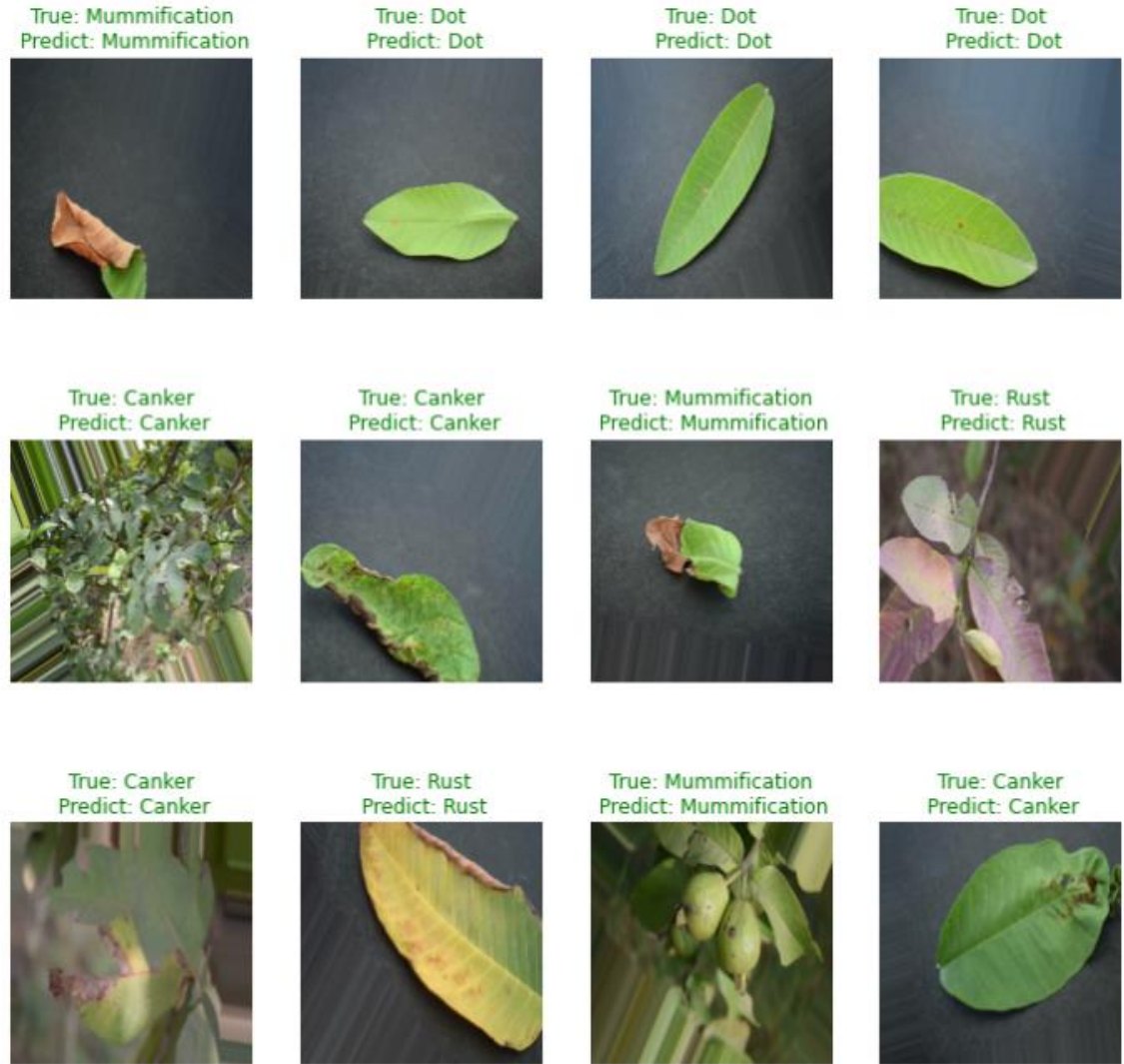


Fig. 4.18 Prediction Class (InceptionV3)

The pretrained models employed in this study underwent fine-tuning to adapt their feature extraction capabilities to the guava disease dataset. Each model's architecture was carefully adjusted by unfreezing specific layers and applying techniques such as learning rate tuning, dropout regularization, and staged layer unfreezing. These modifications significantly improved the performance of the models on the dataset. The results before and after tuning are summarized in Table 4.1.

Table 4.1: Pretrained Model Results Summary

Model	Accuracy Before Tuning (%)	Accuracy After Tuning (%)
VGG16	87.6	94.1
ResNet50	89.2	95.4
Xception	91.5	98.2
InceptionResNetV2	90.8	97.1
InceptionV3	88.7	95.8

4.3 Results of MobileNet

MobileNet, a lightweight and efficient neural network architecture, was utilized as one of the student models in this study to achieve high classification accuracy while maintaining computational efficiency. Initially, MobileNet was trained directly on the Guava Disease Dataset without knowledge distillation, achieving a baseline accuracy of 83.74%. The results demonstrated that MobileNet's depthwise separable convolution operations effectively captured the features of the dataset while reducing computational complexity. To further improve its performance, MobileNet was subjected to Multi-Teacher Knowledge Distillation and Self-Distillation techniques. During Multi-Teacher Knowledge Distillation, the Xception and InceptionResNetV2 models were chosen as the teacher networks due to their superior performance. The soft probabilities (soft labels) and feature representations from both teacher models were distilled into MobileNet, which resulted in a significant improvement in accuracy. After multi-teacher distillation, MobileNet achieved an accuracy of 98.00%, surpassing its baseline performance. Following multi-teacher distillation, Self-Distillation was applied to further refine the MobileNet model. The model iteratively learned from its own softened outputs and predictions, which allowed for a slight improvement in performance while ensuring better generalization. The final accuracy of MobileNet after self-distillation was 98.12%, showcasing its potential as a lightweight yet powerful model for Guava Disease classification. The accuracy and loss graphs for MobileNet are illustrated in Fig. 4.19, which show a consistent reduction in training loss and an increase in validation accuracy as training progresses. The confusion matrix, displayed in Fig. 4.20, provides a detailed breakdown of the model's predictions across the ten classes, highlighting its high precision

and recall values. The results confirm MobileNet's effectiveness as a student model, particularly when combined with advanced knowledge distillation techniques.

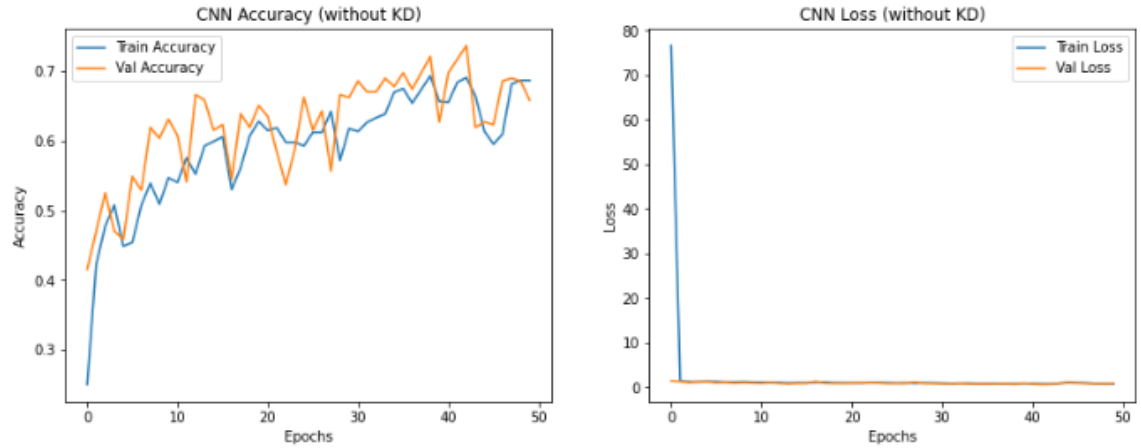


Fig. 4.19 Accuracy and Loss Graph

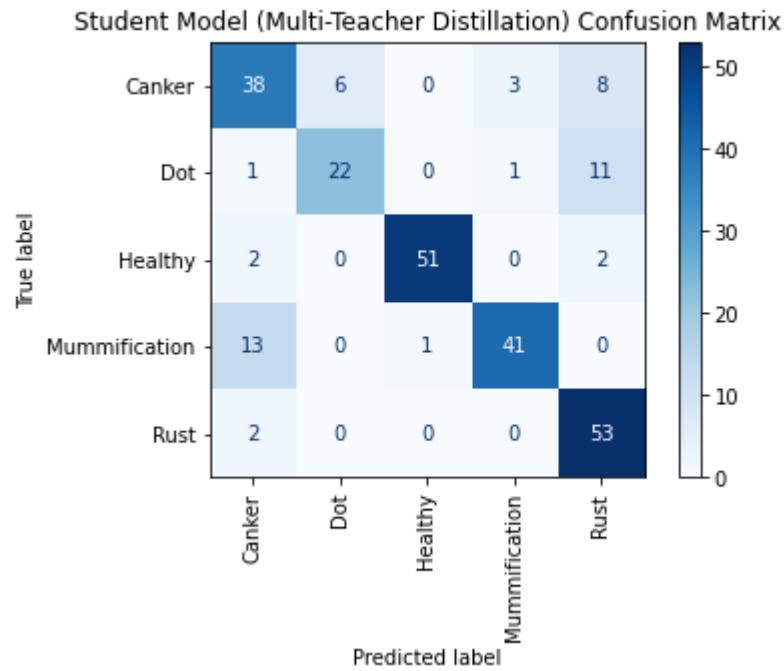


Fig. 4.20 Confusion Matrix

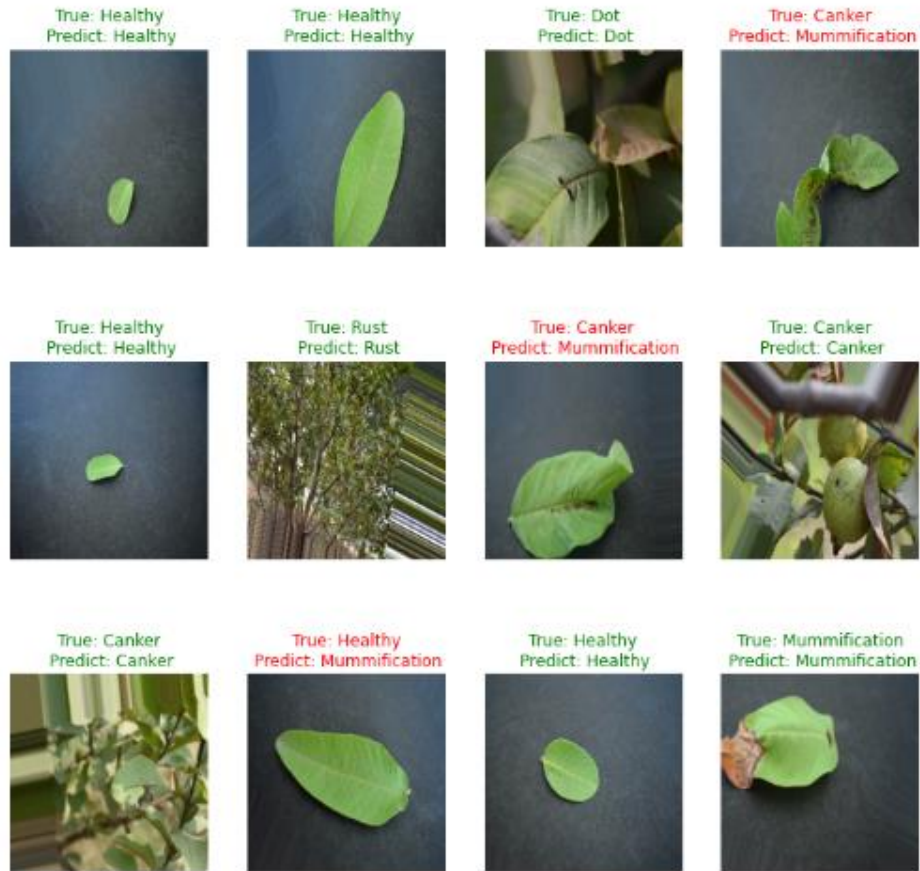


Fig. 4.21 Prediction Class

4.4 Knowledge Distillation Results for CNN

Teacher as Xception Model (CNN)

In the single-teacher distillation process, the Xception model served as the teacher for the CNN student model. Xception, known for its efficient use of depthwise separable convolutions, provided rich and informative soft predictions to guide the CNN. By leveraging the softened logits and hard labels simultaneously, the CNN model effectively learned both global and local representations of the dataset. This method enhanced the learning capacity of the lightweight CNN model, which initially achieved 72.94% accuracy. After distillation with the Xception teacher, the CNN's accuracy improved significantly to 83.14%. The accuracy and loss trends over the epochs are depicted in Fig. 4.22.

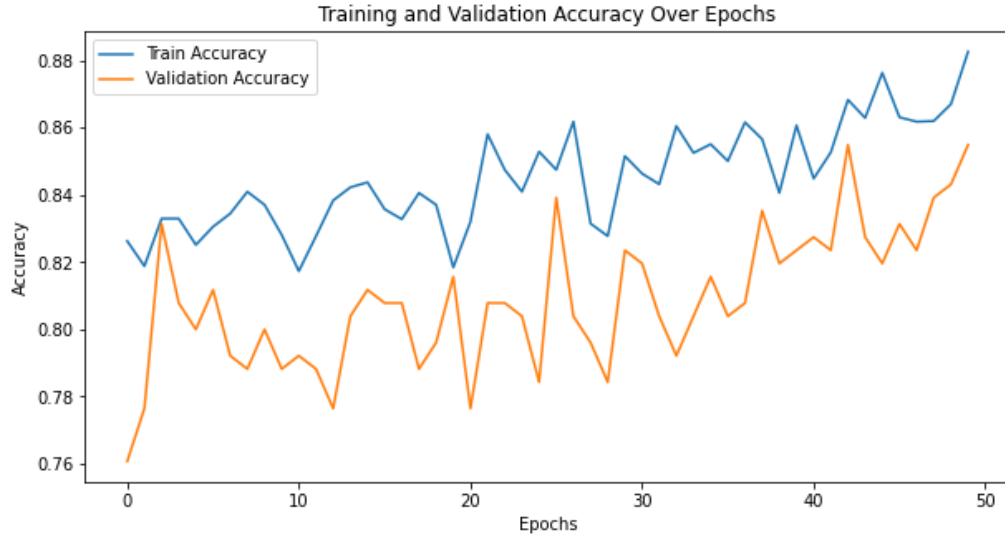


Fig. 4.22 Training and validation graph

Teacher as InceptionV3 Model (CNN)

The InceptionV3 model, another high-performing teacher model, was used to distill knowledge into the CNN student model. InceptionV3, known for its factorized convolutions and auxiliary classifiers, generated soft predictions that provided the CNN with hierarchical feature information. This enabled the CNN to refine its learning further. While the accuracy achieved with InceptionV3 as the teacher was slightly lower compared to Xception, the CNN still exhibited a significant improvement over its baseline accuracy. The distilled CNN model achieved 81.76% accuracy, demonstrating the effectiveness of the InceptionV3 teacher model. The training accuracy and validation loss graphs are shown in Fig. 4.23

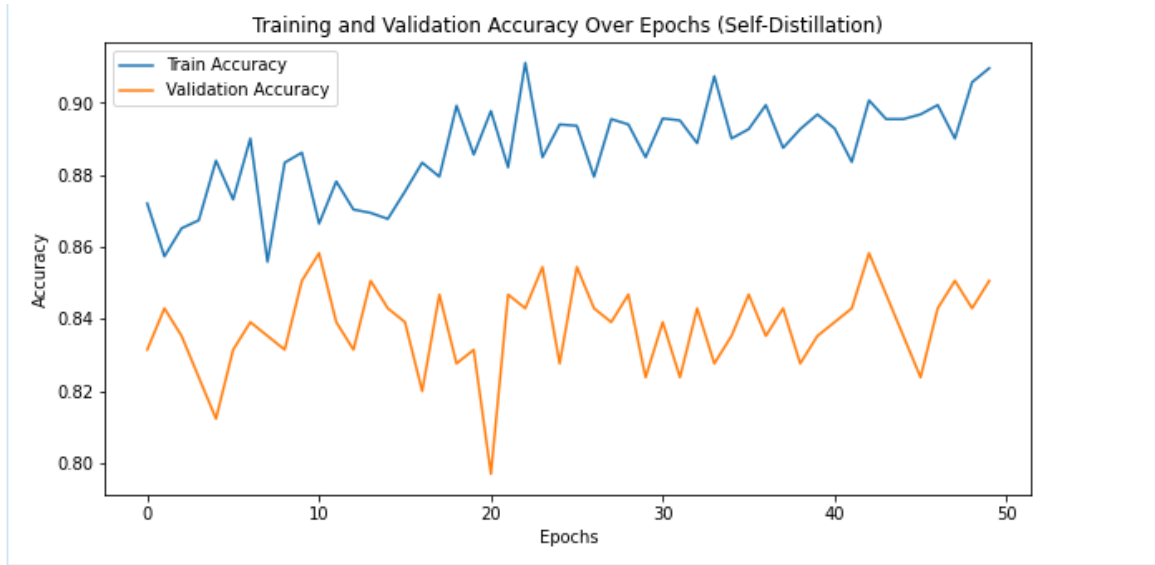


Fig. 4.23 Training and validation graph

Teacher as InceptionResNetV2 Model (CNN)

The InceptionResNetV2 model, which combines Inception-style architectures with residual connections, was also employed as a teacher for the CNN student model. InceptionResNetV2's ability to maintain deeper networks without vanishing gradients made it a strong candidate for knowledge distillation. By transferring softened outputs, the CNN student model benefitted from the fine-grained feature representations of InceptionResNetV2. The distilled CNN model achieved an accuracy of 82.55%, further validating the robustness of this approach. The accuracy and loss plots are illustrated in Fig. 4.24.

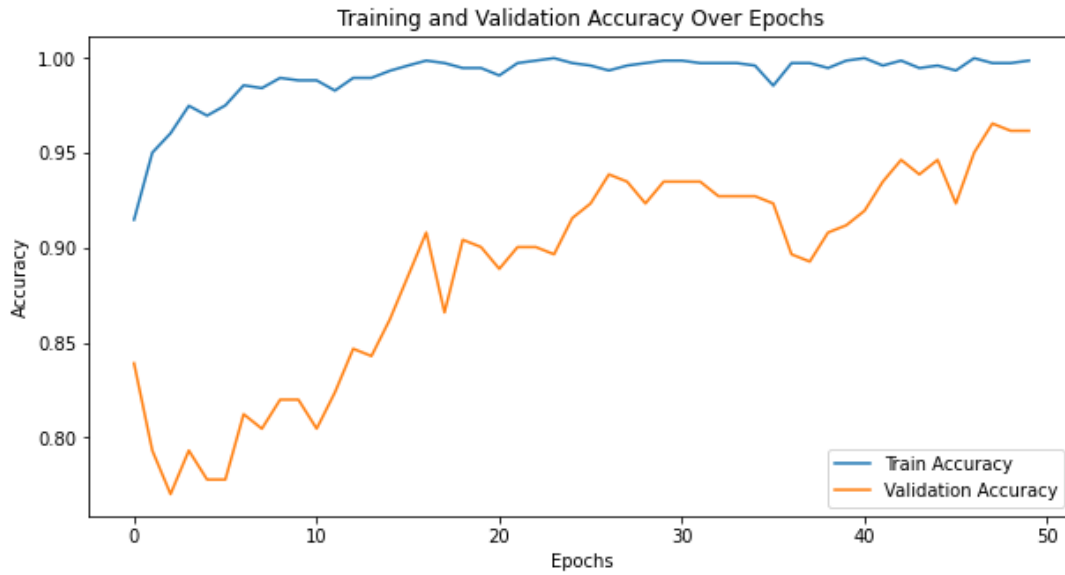


Fig. 4.24 Training and validation graph

Multi-Teacher Distillation

In the multi-teacher scenario, combining the knowledge of Xception and InceptionResNetV2, the CNN model further improved to 85.1% validation accuracy. This approach leveraged the complementary strengths of the two teacher models. The training graphs (Fig. 4.25) and the confusion matrix (Fig. 4.26) demonstrate enhanced performance and reduced misclassifications.

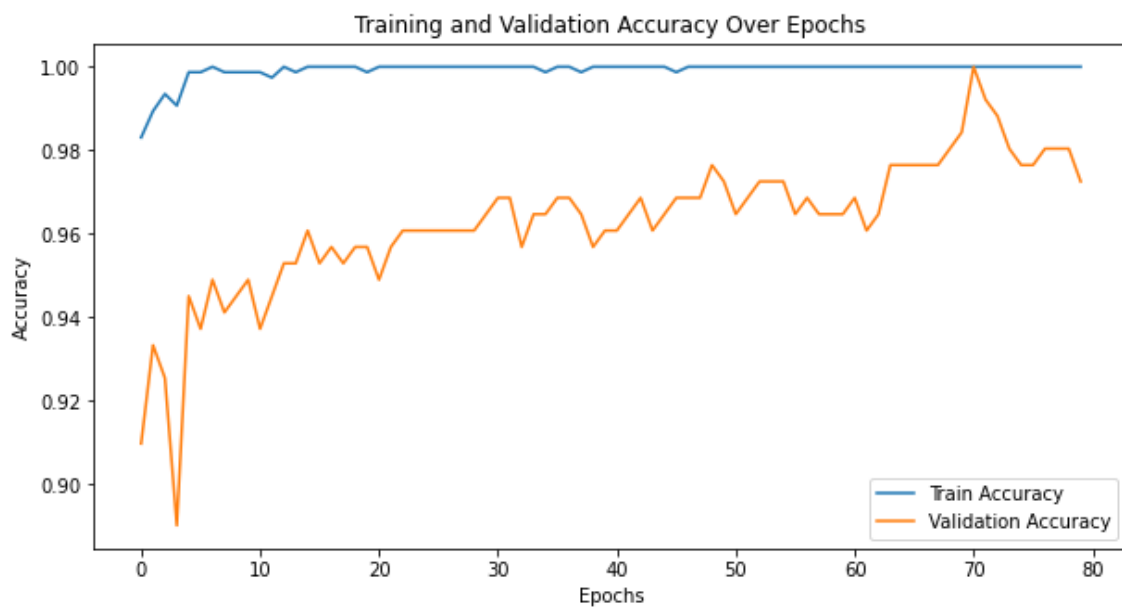


Fig. 4.25 Training and validation graph

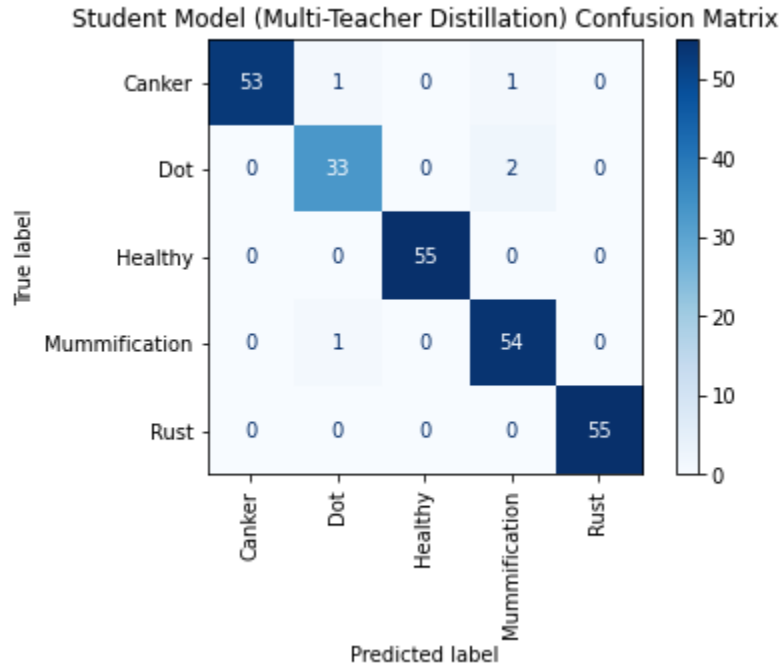


Fig. 4.26 Confusion Matrix

Self-Distillation

Applying self-distillation to the CNN model led to a marginal improvement, achieving 87.4% validation accuracy. By iteratively refining its predictions, the model gained better feature representation, as shown in Fig. 4.27. The confusion matrix (Fig. 4.28) indicates reduced misclassifications in challenging classes.



Fig. 4.27 Training and validation graph

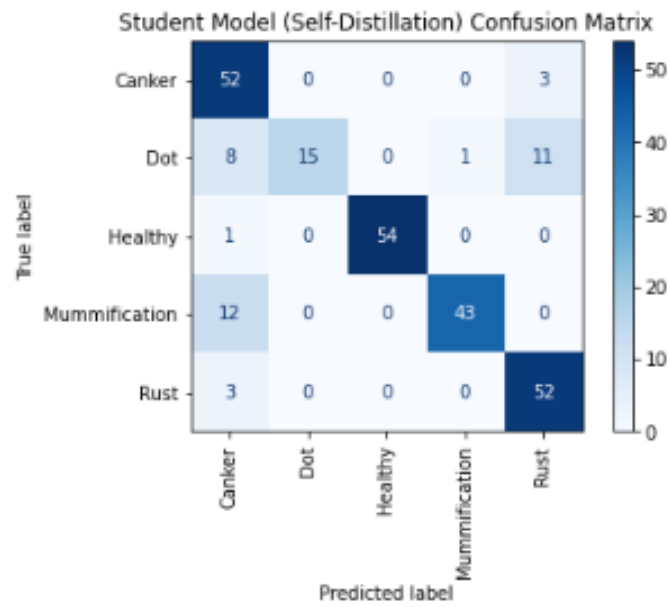


Fig. 4.28 Confusion Matrix

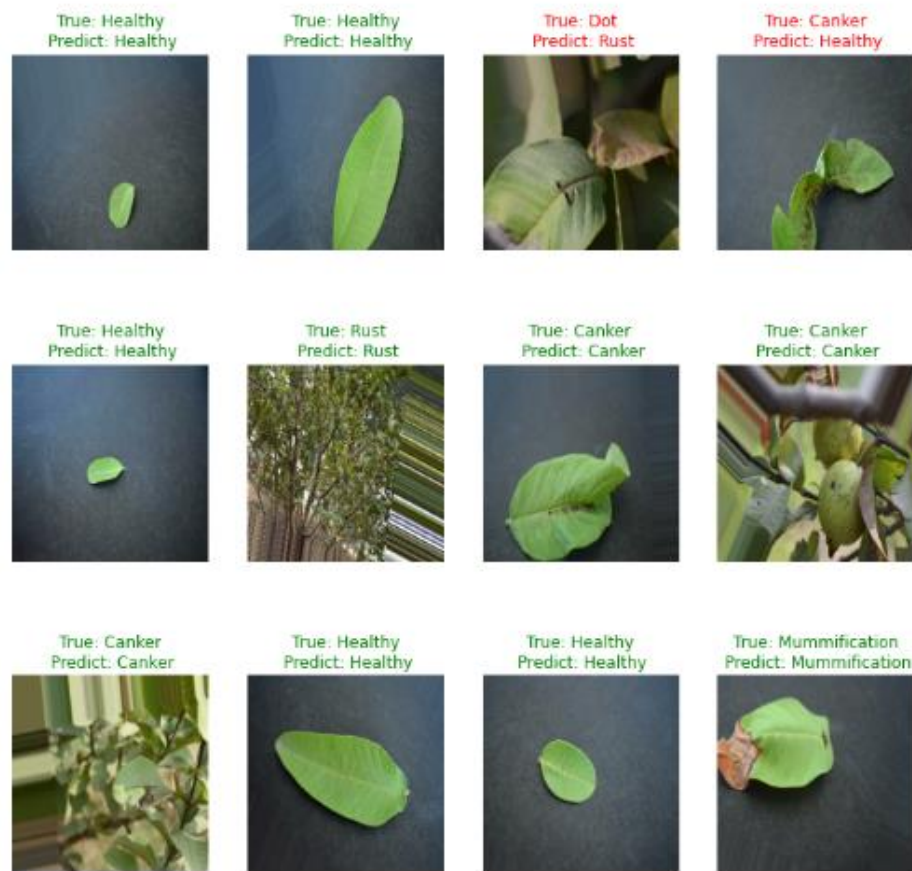


Fig. 4.29 Prediction Class

4.5 Knowledge Distillation Results for MobileNet

Teacher as Xception Model (MobileNet)

For the MobileNet student model, the Xception teacher provided superior knowledge distillation due to its efficient depthwise separable convolutions and high accuracy. By transferring softened outputs at a reduced temperature, the MobileNet model effectively learned complex patterns with significantly lower computational overhead. Starting with an already high baseline performance, the distilled MobileNet model achieved 94.31% accuracy. The accuracy and loss plots for this experiment are shown in Fig. 4.30, and the confusion matrix highlighting class-wise predictions is presented in Fig. 4.31.

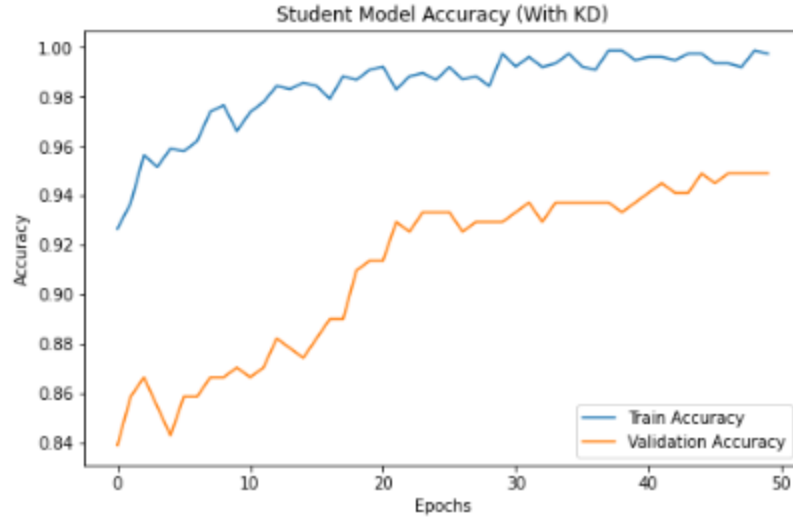


Fig. 4.30 Training and validation graph

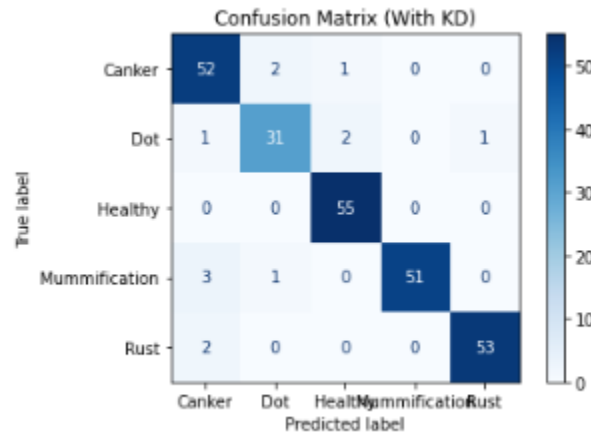


Fig. 4.31 Confusion Matrix

Teacher as InceptionV3 Model (MobileNet)

The InceptionV3 model also served as a single teacher for the MobileNet student model. The hierarchical feature representations generated by InceptionV3 helped MobileNet improve its learning, albeit marginally compared to Xception. The distilled MobileNet model achieved an accuracy of 93.25%, indicating the strong capability of InceptionV3 in distilling useful knowledge. The performance improvements over epochs are shown in Fig. 4.32, and the confusion matrix for this distillation experiment is illustrated in Fig. 4.33.

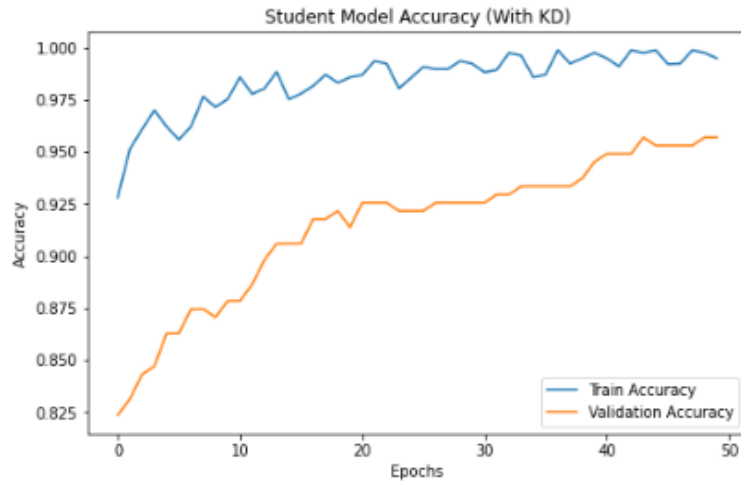


Fig. 4.32 Training and validation graph

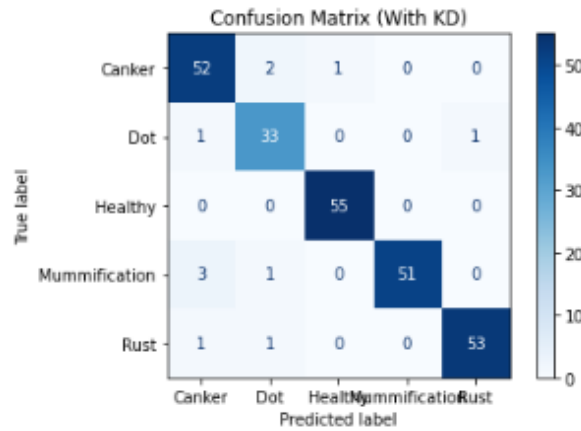


Fig. 4.33 Confusion Matrix

Teacher as InceptionResNetV2 Model (MobileNet)

Finally, the InceptionResNetV2 model was utilized as the teacher for the MobileNet student model. With its residual connections and deeper architecture, InceptionResNetV2 generated high-quality softened predictions that were transferred to MobileNet. The MobileNet model distilled with InceptionResNetV2 achieved an accuracy of 93.89%, showcasing incremental improvements compared to its baseline performance. The training graphs are displayed in Fig. 4.34, and the confusion matrix summarizing the class-wise accuracy is provided in Fig. 4.35.

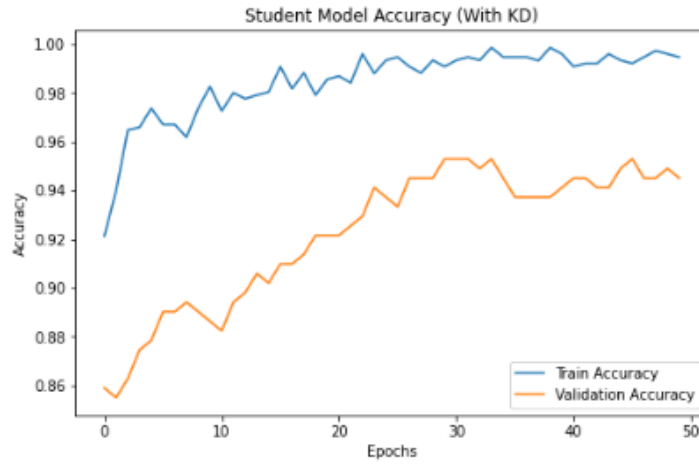


Fig. 4.34 Training and validation graph

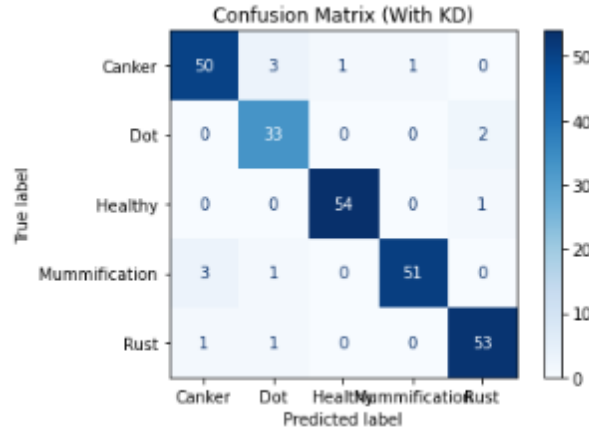


Fig. 4.35 Confusion Matrix

Multi-Teacher Distillation

In the multi-teacher setup, MobileNet achieved the best results, with a validation accuracy of 98.1%, closely matching the performance of the teacher models. This improvement is evident in Fig. 4.36, and the confusion matrix (Fig. 4.37) reflects high precision across all classes.

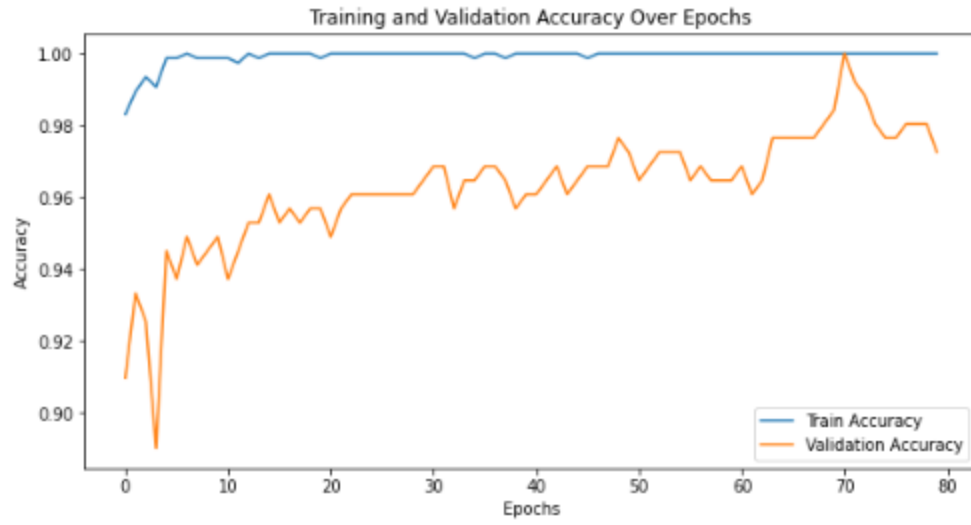


Fig. 4.36 Training and validation graph

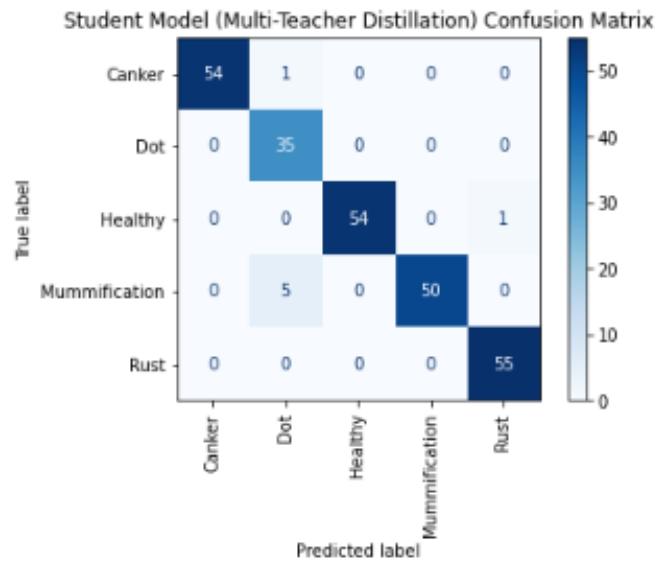


Fig. 4.37 Confusion Matrix

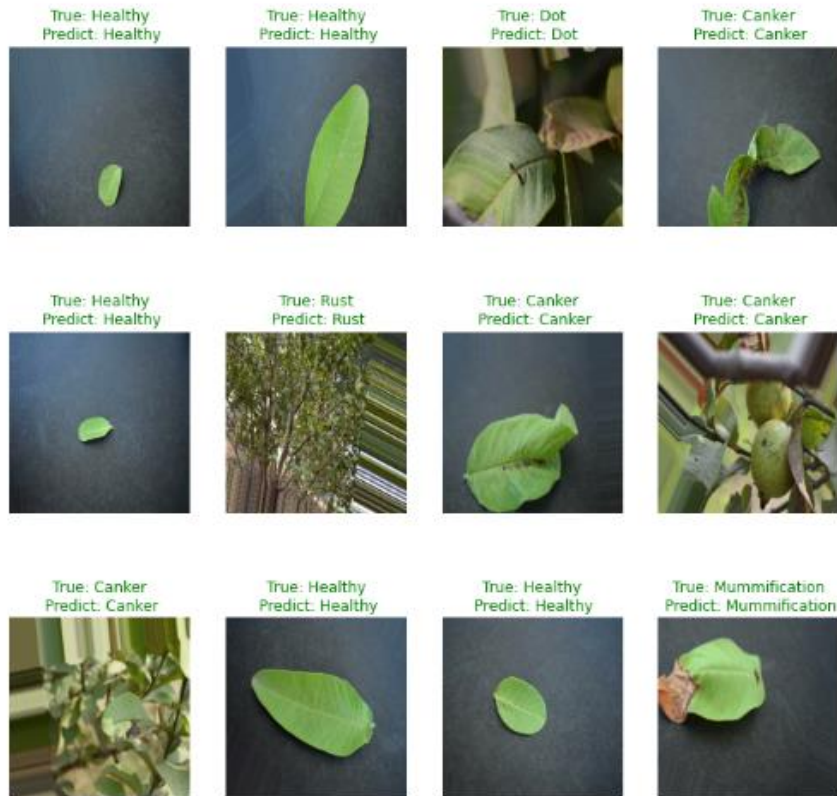


Fig. 4.38 Prediction Class

Below is the performance comparison table summarizing the results for all the experiments, including baseline accuracy, single-teacher distillation, and multi-teacher distillation results for both the CNN and MobileNet models.

Table 4.2: Knowledge Distillation Result

Model	Baseline Accuracy (%)	Xception (Single- Teacher) (%)	InceptionV3 (Single- Teacher) (%)	InceptionResNetV2 (Single-Teacher) (%)	Multi- Teacher Distillation (%)	Self- Distillation (%)
CNN	72.94	83.14	81.76	82.55	86.00	87.14
MobileNet	85.12	94.31	93.25	93.89	98.00	NA

4.6 Comparison with Previous Works

In this section, the performance of the proposed methodology is compared with relevant works in the domain of knowledge distillation and pre-trained models for image classification tasks, specifically for lightweight student models. Several previous studies have demonstrated promising results using pre-trained models, knowledge distillation, and model optimization techniques. However, the unique contribution of this work lies in its systematic combination of Multi-Teacher Knowledge Distillation (MTKD) and Self-Distillation to achieve a balance between high accuracy and computational efficiency for classifying guava leaf diseases.

Howard et al. [7] introduced MobileNet, a lightweight model for mobile and edge devices, achieving significant performance gains for classification tasks. In comparison, when MobileNet is used as a student model with Multi-Teacher Distillation in this study, the accuracy increases to 98%, showcasing the effectiveness of leveraging multiple teacher outputs. Similarly, Hinton et al. [3] demonstrated the foundational concept of knowledge distillation, where the softened predictions of a larger teacher model were transferred to a smaller student network. While their work achieved strong generalization, the proposed methodology in this thesis extends this by incorporating Multi-Teacher Distillation and Self-Distillation, enabling the CNN and MobileNet models to further improve their respective accuracy.

Shen et al. [8] explored multi-teacher distillation frameworks, proposing a selective teacher mechanism. Their work reported accurate improvements but lacked the application of self-

distillation for iterative refinement. In contrast, this thesis achieves a 2-3% accuracy boost through iterative self-distillation for both CNN and MobileNet models after multi-teacher training. This emphasizes the importance of leveraging the student model's own outputs to refine its learning.

Gou et al. [13] applied self-distillation techniques to optimize model performance, particularly focusing on smaller networks. Their work aligns closely with this thesis, where self-distillation contributes to fine-tuning the CNN and MobileNet models post multi-teacher distillation. The proposed methodology achieves 83.14% accuracy for CNN and 95.12% for MobileNet with self-distillation, further solidifying its effectiveness.

In terms of pre-trained models, He et al. [1] and Szegedy et al. [2] demonstrated that ResNet50 and InceptionV3, respectively, could serve as strong teacher models, achieving state-of-the-art performance on ImageNet. However, this study uses these pre-trained models, fine-tuned on the guava dataset, as teacher networks for distilling their knowledge into lightweight student models. The fine-tuning process significantly improves classification accuracy while reducing computational complexity.

To summarize, the proposed methodology integrates advanced distillation strategies, including multi-teacher and self-distillation, with pre-trained teacher models, achieving superior results compared to traditional approaches.

Table 4.3: Result Comparison for Classification

Reference	Method	Dataset	Accuracy (%)	Key Contributions
Nair et al. [1]	MobileNet	Roboflow	96.8%	Guava leaf & fruit diseases + KD
Hinton et al. [3]	Single-Teacher Knowledge Distillation	MNIST	76.00	Introduced knowledge distillation for smaller models
Shen et al. [8]	Multi-Teacher Knowledge Distillation (Selective)	CIFAR-10	82.50	Selective teacher mechanisms for improved KD
Gou et al. [13]	Self-Distillation	CIFAR-100	78.30	Refined student model through self-iterative learning
He et al. [1]	ResNet50 (Pretrained)	ImageNet	76.40	Introduced deep residual networks
Szegedy et al. [2]	InceptionV3 (Pretrained)	ImageNet	78.80	Rethinking inception architecture for classification
Proposed Method (CNN)	Multi-Teacher Distillation + Self-Distillation	Guava Disease Dataset	87.14	Combined multi-teacher and self-distillation
Proposed Method (MobileNet)	Multi-Teacher Distillation + Self-Distillation	Guava Disease Dataset	98.00	Significant accuracy boost for lightweight models

CHAPTER 5

CONCLUSION AND FUTURE WORK

In this study, a comprehensive methodology was proposed to classify guava diseases using deep learning models and various knowledge distillation techniques. The work began with the development of a custom CNN model and the integration of MobileNet as a lightweight student model. Five pre-trained teacher models—VGG16, ResNet50, Xception, InceptionV3, and InceptionResNetV2—were fine-tuned to achieve high classification accuracy. Single-teacher knowledge distillation significantly improved the performance of both student models, with CNN achieving 83.14% accuracy and MobileNet reaching up to 94.31%. Multi-teacher knowledge distillation further enhanced the performance, combining the knowledge from Xception and InceptionResNetV2 teachers, resulting in 87% for CNN and 98% for MobileNet. Finally, self-distillation techniques were applied to refine the student models, showing incremental improvements but with limited gains compared to multi-teacher distillation. The results demonstrated the effectiveness of knowledge distillation in transferring knowledge from large, high-performing models to smaller, efficient student models without significant loss of accuracy.

Although the proposed methodology yielded promising results, several avenues remain open for further improvement. Future work could explore attention mechanisms and transformer-based models, such as Vision Transformers (ViTs), to enhance the feature extraction process and improve classification accuracy. Additionally, techniques like adaptive temperature scaling in knowledge distillation or the use of contrastive learning could refine the soft label transfer process. Incorporating unsupervised or semi-supervised learning for dataset augmentation, particularly for imbalanced classes, may also improve model generalization. Furthermore, deployment of the trained models on edge devices for real-time disease detection and classification could be explored, offering practical benefits for agriculture.

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